Totem-implied repo

Controlling the front-office repo and detecting bad curves

Methodology note • Independent price verification, equity derivatives

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Summary

The problem. Independent price verification implies the repo curve from Totem: the forwards come from consensus call and put prices, the dividends from a separate Totem service. The two services do not have the same contributors, and their number changes with maturity; in our example, 16 option contributors and 12 dividend contributors at the short end, 6 and 4 at five years. The implied repo curve inherits the noise of both panels: it sometimes looks irregular, or jumps from one maturity to the next for no market reason. Yet this is the curve that, through the sensitivities, controls the front-office (FO) repo positions.

The three goals of the note.

G1 *Understand*: where do these shapes come from, and which ones are just normal noise?

G2 *Control*: "I have the forward control, I have the dividend control: how much is the repo control?" Can the repo position be controlled without depending on the dividend contributors?

G3 *Detect*: how can a bad Totem curve be flagged automatically, its cause named, and the flag kept apart from an FO mismark?

The answers.

A1 The implied repo equals the average repo of the option panel plus a *panel gap*: the difference between two dividend averages taken over different contributors. For a European stock, this gap is commonly 4.4 bp at one year and 21 bp at five years (Section 3). No method that only sees Totem data can remove it; one can, however, compute its size and recognise, by their shape, the mechanisms that cause real anomalies.

A2 Repo line = forward line − dividend line + non-forward line. For a book whose value depends on the forwards only (European options, forwards, futures), the non-forward line is zero: the total control does not depend on the dividend contributors, and it is exactly independent of them under full revaluation (Section 4). An exposure vector computed from the FO sensitivities alone shows where the split between repo and dividends matters.

A3 The structural detector judges the Totem–FO difference "in carry", against a noise band computed from the dispersions and contributor counts that Totem publishes, and tests one typical shape per mechanism (Section 5). Its false alarm rate is fixed (2% of clean names); every alarm comes with an explanation in natural units and a repaired curve. On a synthetic benchmark of 5400 names (Section 6), it detects on average as much as the best alternative tested, B-spline plus Gaussian mixture (73% against 73%), finds forward outliers far better (97% against 78%), and does not mistake an FO mismark for a bad Totem curve (3% alarms against 75%).

How to read this note

Each section opens with its goal and closes with a "Key points" box. One synthetic name, the *running example*, illustrates the note from start to finish (green boxes): a European stock seen first in a clean market, then under eight different anomalies. Short proofs follow the statements; the longer ones are in Appendix A.

Question	Section	What you will find
What are we after, and why is it hard?	1	The control chain, the $1/T$ magnifier, the running example
Which notation?	2	Forward, carry difference, Totem data, from Totem to the implied repo
G1 Where do bad curves come from?	3	The panel gap and its size, contributor drop-offs, six mechanisms and their signatures, what the data cannot tell
G2 How to control the repo?	4	The control identity, three books, how much the repo/dividend split matters
G3 How to detect?	5	The noise model, the typical shapes, the test, the calibration, the diagnosis of the running example
Does it work? And B-splines, mixtures, neural networks?	6	The benchmark and the comparison with the starting ideas
What to do in practice?	7	Recommendations, data needed, monthly procedure

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1 The problem and the goal

> **Goal of the section.** Describe the chain from the Totem contributions to the control lines, show why an implied repo curve often looks bad without being bad, and state what the note must deliver.

1.1 The control chain

For each underlying and each control date, the controller has three sources (Fig. 1).

- **Totem options**: for each maturity T_k, consensus call and put prices on a strike grid, the number of contributors n_k^{opt} and the dispersion of their quotes.
- **Totem dividends**: for each period p (usually a calendar year), the consensus level ℓ_p^{tot}, that is, the sum of the dividends going ex in the period, the number of contributors n_p^{div} and their dispersion.
- **The front office**: the spot, the rate curve, the repo curve q^{fo}, the dividend forecast (ex-dates and amounts) and the sensitivities of each book to the repo and to the dividends.

Put–call parity gives the Totem forwards F_k^{tot}. The dividend levels are spread over the FO ex-dates. A bootstrap then gives the repo curve q^{tot} which, with these dividends, reprices the Totem forwards exactly. This curve has two uses: computing the control lines, that is, the Totem minus FO differences times the sensitivities (Section 4), and checking the quality of the curve itself (Section 5).

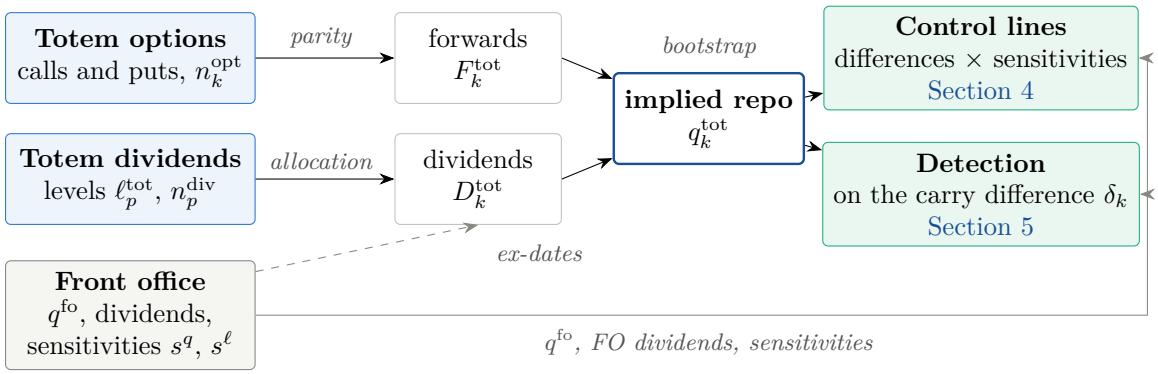


Figure 1: The control chain. The two Totem services have different contributors, and their number changes with maturity; the implied repo curve q^{tot} inherits the noise of both.

1.2 Why the curve looks bad

Two reasons, neither of them an error, are enough to make a clean curve look irregular.

Two panels, two averages. The Totem forward is the average of the option panel's forwards, and each contributor's forward contains its own view of the dividends. The controller takes out the dividends of another panel, the dividend service's. If the two panels do not share the same average view of the dividends, the difference ends up in the implied repo. This is not a Totem error: it is the mechanical consequence of two averages taken over different contributors. Its size depends on the contributor counts, which fall with maturity, and on the dispersion of the dividend views, which grows with it (Section 3.1).

The $1/T$ magnifier. The repo is a rate: an error ε on the log of the forward becomes an error ε/T on the repo. An error of 0.1% on the forward is worth 120 bp of repo at one month and 2 bp at five years. The short end therefore always looks noisy, while a large error at the long end goes unnoticed.

A curve that looks irregular is therefore not necessarily bad, and a curve that looks smooth is not necessarily good. The rest of the note works on the *carry difference* $\delta_k = (q_k^{\text{tot}} - q_k^{\text{fo}})\,T_k$, where errors add up without being magnified, and compares it with a noise band computed from the statistics that Totem publishes.

1.3 The running example

Running example

A synthetic European stock (Section 6): spot $S_0 = 100$, an annual dividend of about 4.2 going ex in early May (the first on 4 May 2027), an FO repo going from 10 bp at the short end to 5 bp at five years. Totem has 16 option contributors on the short maturities and 6 at five years; the dividend service has 12 on 2027 and 4 on 2031. The eleven Totem maturities run from one month to five years. Figure 2 shows the clean market; the same name is taken up again, with the same contributor noise, under eight anomalies in Section 3.3 (Fig. 4) and run through the detector in Section 5.7.

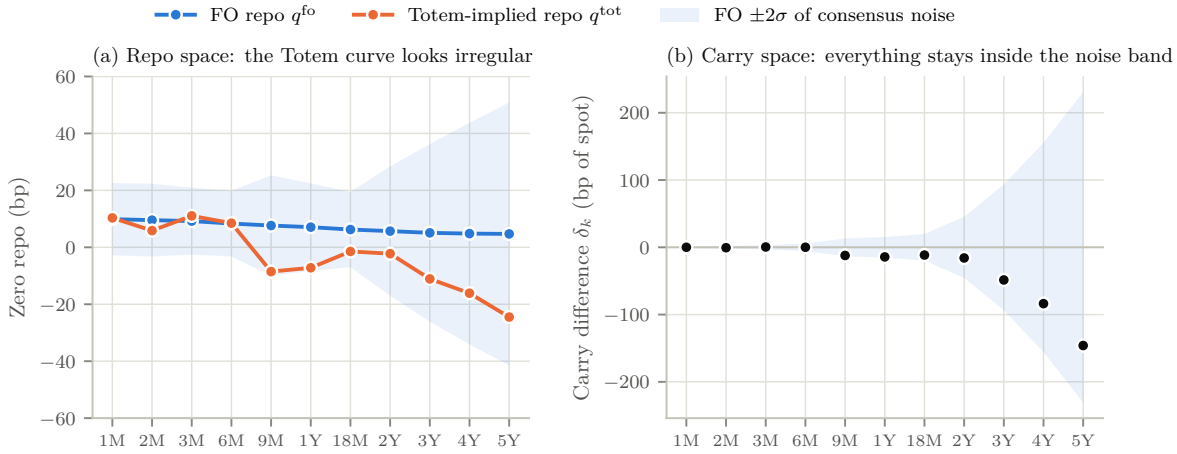


Figure 2: Running example, clean market. (a) FO repo and Totem-implied repo, with the band of ± 2 standard deviations of consensus noise around the FO repo. (b) The same difference seen in carry: each point is $\delta_k = (q_k^{\text{tot}} - q_k^{\text{fo}})\,T_k$, in basis points of spot.

In repo space, the Totem curve looks badly shaped: it loses 17 bp between six and nine months and ends 29 bp below the FO at five years. In carry space, every point stays inside the band. The drop at nine months corresponds to a gap of 0.12 on the May 2027 dividend between the option panel's view and the dividend panel's, that is, 3% of the dividend and 1.8 standard deviations; at five years, the carry difference is −146 bp of spot for a standard deviation of 115 bp. The detector of Section 5 flags nothing. This curve is clean: it is what an implied repo from two different panels normally looks like.

1.4 What the note must deliver

The three goals of the first page translate into four deliverables.

1. **A diagnosis** (G1, Section 3): a decomposition of the implied repo that separates the expected noise, whose size can be computed, from the mechanisms that cause real

anomalies, each with a recognisable shape.

2. **A control rule** (G2, Section 4): a decomposition of the control lines that does not depend on the dividend contributors where possible, and a measure, computed from the sensitivities, of what it still leaves depending on them.
3. **A detector** (G3, Section 5) with a known false alarm rate, which names the mechanism at work, gives its size in natural units (spot off by so much, May 2027 dividend too low by so much), repairs the curve, and tells a Totem anomaly from an FO mismark.
4. **A comparison** (Section 6) with the starting ideas: B-spline coefficients by region and category clustered by a Gaussian mixture, small neural networks specialised by anomaly type.

Key points

- The implied repo curve is a residual: one panel's forwards minus another panel's dividends. It inherits the noise of both.
- The eye misleads: the repo magnifies short-end errors by $1/T$ and flattens long-end ones. Judge the curve on the carry difference δ_k, against a noise band.
- In the running example, a curve that loses 17 bp in three months and ends 29 bp below the FO is perfectly clean.

2 Setup and notation

Goal of the section. Fix the notation and establish the three tools the rest of the note needs: the forward formula, the carry difference, and the passage from Totem data to the implied repo.

2.1 Conventions

Time is measured in years from the valuation date, and rates are continuously compounded zero rates. Scalars and vectors are written in lower case, matrices in upper case; $A^\top$ is the transpose of A, $\mathcal{M}_{N,m}(\mathbb{R})$ the space of matrices with N rows and m columns, $\langle x, y\rangle = x^\top y$ the inner product of $\mathbb{R}^N$ and $\|x\|$ the associated norm. The superscript tot marks a quantity from Totem and the superscript fo a front-office quantity. A basis point (bp) is 10^{-4}; a carry difference is expressed in basis points of spot. The notation is summarised in Appendix D.

2.2 Forward and discrete dividends

Let S_0 be the spot, $r(T)$ the zero rate of maturity T, and $q(T)$ the zero *repo* rate, that is, the cost of borrowing the stock: holding the stock financed by a repo earns $r - q$ per unit of time. Dividends are discrete: an amount d_i goes ex at time t_i. Let $\tilde{d}_i = d_i\, e^{-(r(t_i)-q(t_i))\,t_i}$ be the dividend discounted at the carry rate.

Definition 2.1 (Forward)**.** The forward of maturity T is

$$F(T) = \big(S_0 - D(T)\big)\, e^{(r(T)-q(T))\,T}, \qquad D(T) = \sum_{t_i \le T} \tilde{d}_i. \tag{1}$$

$D(T)$ is the sum of the dividends going ex before T, discounted at the carry rate.

This is the usual replication argument: buy the stock financed at the rate $r - q$ and carry the

dividends received forward. The repo also enters D, weakly, through the discounting of the dividends; the bootstrap of Section 2.5 handles this coupling exactly.

Lemma 2.2 (Log of the forward)**.** $\ln F(T) = \big(r(T) - q(T)\big)\,T + \ln\big(S_0 - D(T)\big)$. *Hence, to first order, perturbations of the forward, the dividends and the repo satisfy*

$$\delta\big(q(T)\,T\big) = \delta\big(r(T)\,T\big) - \delta \ln F(T) - \frac{\delta D(T)}{S_0 - D(T)}. \tag{2}$$

Proof. Take the log of (1) and differentiate, with $\delta \ln(S_0 - D) = -\delta D/(S_0 - D) + O(\delta D^2)$. □

2.3 The carry difference

The Totem maturities, or *pillars*, are $T_1 < \cdots < T_N$ (here $N = 11$, from one month to five years). For any quantity x, we write $x_k = x(T_k)$.

Definition 2.3 (Carry difference)**.** The *carry difference* at pillar T_k is

$$\delta_k = \big(q_k^{\text{tot}} - q_k^{\text{fo}}\big)\,T_k, \qquad k \in \{1, \dots, N\}, \tag{3}$$

where q^{tot} is the Totem-implied repo (Definition 2.11) and q^{fo} the FO repo. We write $\delta = (\delta_1, \dots, \delta_N) \in \mathbb{R}^N$.

Proposition 2.4 (The repo, a difference of forward and dividends)**.** *The controller uses the same rate curve on both sides. Then, up to the coupling through the discounting of the dividends,*

$$\delta_k = -\ln \frac{F_k^{\text{tot}}}{F_k^{\text{fo}}} + \ln \frac{S_0 - D_k^{\text{tot}}}{S_0 - D_k^{\text{fo}}} \;\approx\; -\frac{F_k^{\text{tot}} - F_k^{\text{fo}}}{F_k} - \frac{D_k^{\text{tot}} - D_k^{\text{fo}}}{S_0 - D_k}. \tag{4}$$

Proof. By construction, the Totem-implied repo reprices the Totem forward with the Totem dividends, and the FO repo the FO forward with the FO dividends. Apply Lemma 2.2 on both sides and subtract: the rate terms cancel. The approximation comes from $\ln(1 + x) = x + O(x^2)$. □

In carry terms, the repo difference is therefore the relative forward difference minus the relative dividend difference. This is already, in elementary form, the controller's intuition: repo control = forward control − dividend control. Section 4 turns it into an identity on the control lines.

Key points

Why carry rather than repo: in δ, the effects of the different mechanisms *add up*, and nothing is magnified. A constant error ε on $\ln F$ is a constant in carry and a hyperbola ε/T in repo. Every test in this note therefore works on δ.

2.4 The Totem data

Assumption 2.5 (Consistent contributors)**.** Contributor j has a repo view $q_j(\cdot)$ and a dividend view $D_j(\cdot)$, and quotes option prices consistent with the forward $F_j(T) = \big(S_0 - D_j(T)\big)\,e^{(r(T) - q_j(T))T}$, the spot and the rate curve being common.

Assumption 2.6 (Panels)**.** At pillar T_k, the option panel $\mathcal{P}_k^{\text{opt}}$ has n_k^{opt} contributors; for period p, the dividend panel $\mathcal{P}_p^{\text{div}}$ has n_p^{div}. Panels are nested in maturity: a contributor who quotes a long maturity also quotes the shorter ones. View errors are independent from one contributor to another. The two services have n^{com} contributors in common ($n^{\text{com}} = 0$ for disjoint panels).

Assumption 2.7 (Consensus)**.** The consensus is the arithmetic mean over the panel (after an optional outlier filter). For each maturity and each period, Totem publishes the consensus, the dispersion across contributors and their number.

2.5 From the consensus to the implied repo

The passage has three steps.

Forwards from put–call parity.

Lemma 2.8 (Parity and consensus forward)**.** *If every contributor quotes European calls and puts with the same discount factor $B(0,T)$, then $C_j(K) - P_j(K) = B(0,T)\,(F_j - K)$ for every strike K. If the consensus averages the same contributors at every strike, $\bar{C}(K) - \bar{P}(K) = B(0,T)\,(\bar{F} - K)$, where $\bar{F}$ is the average of the contributors' forwards. The strike where $\bar{C} = \bar{P}$ is therefore $\bar{F}$, and it does not depend on $B(0,T)$.*

Proof. The first equality is put–call parity; the second follows by averaging the first, which is affine; the third by setting the second to zero. □

In practice, the line $K \mapsto \bar{C}(K) - \bar{P}(K)$ is fitted by weighted least squares; its root is the Totem forward F_k^{tot} and its slope gives an implied discount factor, used as a diagnostic. Two situations break the lemma: an outlier filter applied separately per strike and option type (the averaged contributors are then no longer the same), and American exercise. A worked example and the case of discount factors that differ across contributors are in Appendix A.2.

Dividend allocation. The Totem levels ℓ_p are sums of the dividends going ex in the period $I_p = (e_{p-1}, e_p]$, $p \in \{1, \dots, m\}$ (here $m = 6$: the rest of 2026, then 2027 to 2031). The controller spreads them over the ex-dates of the FO forecast: a profile gives a weight $\pi_i \geq 0$ to each date, with $\sum_{t_i \in I_p} \pi_i = 1$, and the allocated dividend is $d_i = \ell_p\, \pi_i$.

Lemma 2.9 (Allocation)**.** *For a fixed repo,*

$$D_k = \sum_{p=1}^{m} a_{kp}\, \ell_p, \qquad a_{kp} = \sum_{t_i \in I_p,\ t_i \leq T_k} \pi_i\, e^{-(r(t_i) - q(t_i))\, t_i}.$$

We call $A = (a_{kp}) \in \mathcal{M}_{N,m}(\mathbb{R})$ the allocation matrix.

Proof. Replace d_i by $\ell_p \pi_i$ in (1) and group the terms by period. □

The coefficient a_{kp} is the discounted share of period p that goes ex before T_k. It is zero if the period starts after T_k, and equal to the discount factor of the period (close to 1) if the whole period goes ex before T_k; in between, for a period that straddles a pillar, it depends on the profile: this is exactly where the profile matters.

Bootstrap. We look for a repo curve with a constant forward repo between pillars, that is, such that $t \mapsto q(t)\, t$ is affine on each $[T_{k-1}, T_k]$, which reprices the forwards. It is built pillar by pillar: knowing $q_1, \dots, q_{k-1}$, the unknown is $x = q_k T_k$, and the model forward at T_k is a function $\mathcal{F}_k(x)$, written out in Appendix A.1.

Proposition 2.10 (The bootstrap is well posed)**.** *Assume $S_0 - D_{k-1} > 0$. The function $\mathcal{F}_k$ satisfies $\mathcal{F}_k' \leq -\mathcal{F}_k$; it is therefore strictly decreasing where it is positive. For every $F_k > 0$, the equation $\mathcal{F}_k(x) = F_k$ has a unique solution, and Newton's method applied to $x \mapsto \ln \mathcal{F}_k(x) - \ln F_k$, safeguarded by bisection, converges to it.*

The proof, a worked example and the algorithm are in Appendix A.1.

Definition 2.11 (Totem-implied repo)**.** The *Totem-implied repo* q^{tot} is the bootstrap solution for the forwards F_k^{tot} and the Totem dividend levels ℓ_p^{tot} allocated on the FO profile. We also write $\hat{q}$ for the solution with the Totem forwards and the *FO* dividends: it depends on Totem only through its forwards.

Key points

- $\delta_k = (q_k^{\text{tot}} - q_k^{\text{fo}})\,T_k$: the repo difference expressed in carry, in basis points of spot.
- $\delta_k \approx -$ (relative forward difference) $-$ (relative dividend difference): the implied repo is a residual.
- The passage from Totem to the implied repo (parity, allocation, bootstrap) is exact and well posed; the difficulties come from the data, not from the algorithm.

3 Where do bad curves come from?

Goal of the section. (G1) Decompose the implied repo, compute the size of the normal noise, and give the shape that each mechanism causing real anomalies leaves in the carry difference.

3.1 The panel gap

We work under Assumptions 2.5 to 2.7 at a fixed maturity T. The contributors' views are perturbations of a common centre, $q_j = q + u_j$ and $D_j = D + v_j$, where u_j and v_j are of order ϵ. Let $\bar{q}^{\text{opt}}$ and $\bar{D}^{\text{opt}}$ be the averages of the q_j and of the D_j over the option panel, and $\bar{D}^{\text{div}}$ the consensus of the dividend panel, allocated at maturity T.

Proposition 3.1 (Decomposition of the implied repo)**.**

$$q^{\text{tot}}(T) = \bar{q}^{\text{opt}}(T) + \frac{\bar{D}^{\text{opt}}(T) - \bar{D}^{\text{div}}(T)}{T\,(S_0 - D(T))} + O(\epsilon^2). \tag{5}$$

Proof. For $j \in \mathcal{P}^{\text{opt}}$, Lemma 2.2 gives $\ln F_j = \ln F - T\,u_j - v_j/(S_0 - D) + O(\epsilon^2)$. The Totem forward is the average of the F_j, and the log of an average equals the average of the logs up to a term of order ϵ^2; hence $\ln F^{\text{tot}} = \ln F - T\,\bar{u}^{\text{opt}} - \bar{v}^{\text{opt}}/(S_0 - D) + O(\epsilon^2)$. The controller takes out the consensus dividends $D^{\text{tot}} = D + \bar{v}^{\text{div}}$:

$$q^{\text{tot}}T = rT - \ln F^{\text{tot}} + \ln(S_0 - D^{\text{tot}}) = qT + T\,\bar{u}^{\text{opt}} + \frac{\bar{v}^{\text{opt}} - \bar{v}^{\text{div}}}{S_0 - D} + O(\epsilon^2).$$

Divide by T, with $q + \bar{u}^{\text{opt}} = \bar{q}^{\text{opt}}$ and $\bar{v}^{\text{opt}} - \bar{v}^{\text{div}} = \bar{D}^{\text{opt}} - \bar{D}^{\text{div}}$. □

The implied repo is therefore the average repo of the option panel plus a *panel gap*: the difference between the option panel's average dividend view, which nobody observes, and the dividend panel's, divided by $T(S_0 - D)$. For a given contributor, an error on its dividends cancels exactly in its implied repo; between two panels, it does not.

Corollary 3.2 (Size of the panel gap)**.** *Assume the view errors on $D(T)$ are independent and identically distributed with standard deviation $\sigma_D(T)$ in both panels, and the n^{com} contributors common to both services consistent with themselves. Then*

$$\sigma(q^{\text{tot}} - \bar{q}^{\text{opt}}) = \frac{\sigma_D(T)}{T\,(S_0 - D(T))}\sqrt{\frac{1}{n^{\text{opt}}} + \frac{1}{n^{\text{div}}} - \frac{2\,n^{\text{com}}}{n^{\text{opt}} n^{\text{div}}}}. \tag{6}$$

Proof. $\bar{D}^{\text{opt}} - \bar{D}^{\text{div}} = \sum_j (\alpha_j - \gamma_j)\, v_j$, with $\alpha_j = 1/n^{\text{opt}}$ on the option panel, $\gamma_j = 1/n^{\text{div}}$ on the dividend panel, and 0 elsewhere. Its variance is $\sigma_D^2 \sum_j (\alpha_j - \gamma_j)^2 = \sigma_D^2 [n^{\text{com}}(1/n^{\text{opt}} - 1/n^{\text{div}})^2 + (n^{\text{opt}} - n^{\text{com}})/(n^{\text{opt}})^2 + (n^{\text{div}} - n^{\text{com}})/(n^{\text{div}})^2]$, which simplifies to $\sigma_D^2(1/n^{\text{opt}} + 1/n^{\text{div}} - 2n^{\text{com}}/(n^{\text{opt}} n^{\text{div}}))$. □

The gap vanishes if the two panels are identical ($n^{\text{com}} = n^{\text{opt}} = n^{\text{div}}$); it is largest when they are disjoint. The dispersion $\sigma_D(T)$ of a cumulative dividend grows at least like T: the panel gap is small in carry at the short end, but it does not shrink in repo at the long end.

Example 3.3 (Size of the gap for a stock)**.** We take the running example's model: spot 100, an annual dividend of 4.2 going ex in early May, and each contributor wrong on the dividend going ex at t by a relative error with standard deviation $0.01 + 0.03\,t$, part of it common to all its dividends. The panels are disjoint, with the running example's sizes, and the option contributors also disagree by 15 bp on the repo, which the implied repo must not pick up. Formula (6) and a contributor-level simulation give:

Maturity	$D(T)$	$\sigma_D(T)$	n^{opt}	n^{div}	Formula (bp)	Simulation (bp)
9M	4.13	0.11	15	12	5.9	5.8
1Y	4.13	0.11	15	12	4.4	4.4
18M	4.13	0.11	13	10	3.2	3.2
2Y	8.14	0.30	12	10	6.9	7.0
3Y	12.03	0.56	7	7	11.3	11.3
5Y	19.47	1.29	6	4	20.7	20.5

The gap alone produces 4.4 bp of repo noise at one year and 21 bp at five years, with no error at all in Totem. It is zero before the first ex-date, jumps at each ex-date, then decays like $1/T$ until the next one (Fig. 3).

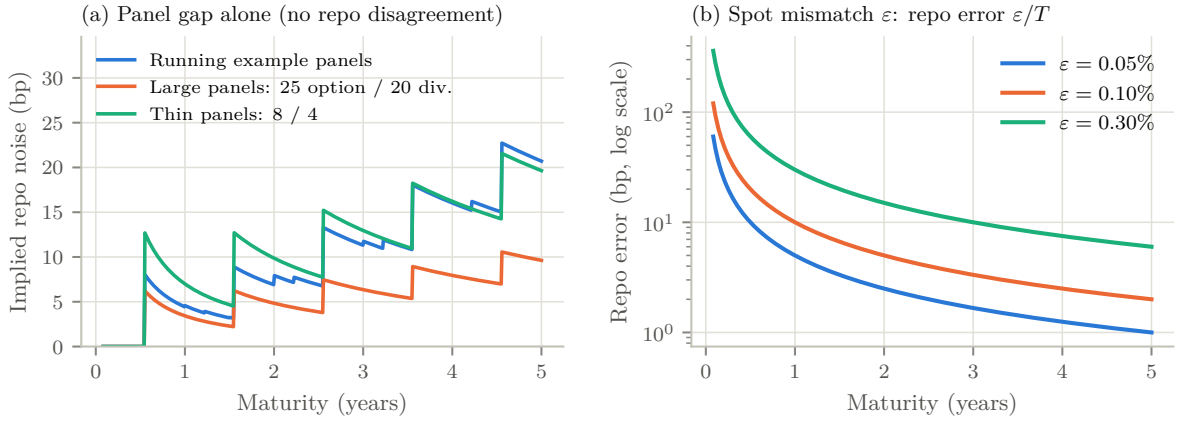


Figure 3: (a) Implied repo noise from the panel gap alone, formula (6), for the running example's panels and two other sizes (options / dividends): it appears at each ex-date, then decays like $1/T$. (b) A spot mismatch ε is constant in carry, hence ε/T in repo (Proposition 3.5).

3.2 Contributor drop-offs

Panels shrink with maturity. Since they are nested, the average over a small panel is not independent of the average over a larger panel that contains it.

Proposition 3.4 (Averages over nested panels)**.** *Let e_{jk} be the error of contributor j at maturity T_k, independent across contributors, with* $\operatorname{Var} e_{jk} = \sigma_k^2$ *and* $\operatorname{Cov}(e_{jk}, e_{jl}) = \rho_{kl}\,\sigma_k \sigma_l$.

If the panels are nested, with sizes n_k, the averages $\bar{e}_k$ satisfy

$$\mathrm{Cov}(\bar{e}_k, \bar{e}_l) = \frac{\rho_{kl}\,\sigma_k\sigma_l}{\max(n_k, n_l)}. \tag{7}$$

In particular, for persistent views ($\rho_{kl} = 1$, $\sigma_k = \sigma_l = \sigma$) and $n_l \le n_k$, the jump of the average when the panel goes from n_k to n_l contributors has standard deviation $\sigma\sqrt{1/n_l - 1/n_k}$.

Proof. The two averages share $\min(n_k, n_l)$ contributors, hence

$$\mathrm{Cov}(\bar{e}_k, \bar{e}_l) = \frac{\min(n_k, n_l)}{n_k n_l}\,\rho_{kl}\sigma_k\sigma_l = \frac{\rho_{kl}\,\sigma_k\sigma_l}{\max(n_k, n_l)}.$$

Then $\mathrm{Var}(\bar{e}_l - \bar{e}_k) = \sigma^2(1/n_l + 1/n_k - 2/n_k) = \sigma^2(1/n_l - 1/n_k)$. □

In units of contributor dispersion, the consensus thus jumps by itself by $0.14\,\sigma$ when the panel goes from 16 to 12 contributors, by $0.24\,\sigma$ from 12 to 7 (the running example's option panel between two and three years), by $0.28\,\sigma$ from 15 to 7 and by $0.39\,\sigma$ from 10 to 4. A jump of that size at a contributor drop-off is noise, not information; the noise model of Section 5 reproduces it through (7).

3.3 Six mechanisms, six signatures

Beyond the noise, some mechanisms cause real anomalies. Each leaves a characteristic shape in the carry difference.

Proposition 3.5 (Signatures)**.** *To first order, each mechanism adds the following contribution to the carry difference δ.*

(i) Spot mismatch. *If the option contributors price off a spot $S_0(1+\varepsilon)$ while the controller uses S_0: $-\varepsilon S_0/(S_0 - D_k) \approx -\varepsilon$ at every pillar, that is, a repo error $-\varepsilon/T_k$. A dividend going ex before T_1 and wrong by δd in the dividend service gives the same shape, $-\delta\tilde{d}/(S_0 - D_k)$.*

(ii) Error on one dividend period. *If the Totem level of period p is wrong by $\delta\ell_p$, the contribution is $-a_{kp}\,\delta\ell_p/(S_0 - D_k)$: a step that starts at the first pillar after the ex-date; in repo, a jump that decays like $1/T$.*

(iii) Ex-date disagreement. *Take a dividend going ex at $t_i \in (T_k, T_{k+1})$, and a fraction f of the option panel that puts it before T_k: $+f\,\tilde{d}_i/(S_0 - D_k)$ at pillar T_k alone, a spike. The variance of the $\ln F_j$ at T_k rises by $f(1-f)\,\tilde{d}_i^2/(S_0 - D_k)^2$.*

(iv) Stale ex-date profile. *If the forwards use the true dates t_i but the allocation puts them at $t_i + \Delta$ with $\Delta > 0$: $+\tilde{d}_i/(S_0 - D_k)$ at the pillars $T_k \in [t_i, t_i + \Delta)$, and nothing elsewhere (for $\Delta < 0$, $-\tilde{d}_i/(S_0 - D_k)$ at the pillars $T_k \in [t_i + \Delta, t_i)$). For a seasonal payer, the affected pillars come back every year.*

(v) Forward outlier. *If the consensus $\ln F_k^{\mathrm{tot}}$ is off by μ: $-\mu$ at pillar T_k alone, a spike.*

(vi) Proportional dividend bias *(for example gross versus net of withholding tax). If every Totem dividend is $(1+\tau)$ times the one in the forwards: $-\tau D_k/(S_0 - D_k)$, a staircase that steps up at every ex-date. For a continuous stream with yield y, $D_k \approx S_0\,y\,T_k$ and the contribution is about $-\tau\,y\,T_k$: exactly the shape of a constant repo difference of $-\tau y$.*

Two more shapes complete the picture: a contributor drop-off at pillar T_k gives a step at T_k (Proposition 3.4), and an FO repo mismark of Δq gives $-\Delta q(T_k)\,T_k$, a smooth repo difference.

Proof. Apply (4) to perturbed forwards or dividends. (i) Each contributor's forward becomes $(S_0(1+\varepsilon) - D)e^{(r-q)T}$, so $\ln F_k^{\text{tot}}$ moves by $\ln(1 + \varepsilon S_0/(S_0 - D_k))$; for the near dividend, D_k^{tot} moves by $\delta\tilde{d}$ at every pillar. (ii) Lemma 2.9 gives $\delta D_k^{\text{tot}} = a_{kp}\,\delta\ell_p$, zero before the ex-date and roughly constant after. (iii) At T_k, a fraction f of the option panel includes the dividend, so $\ln F_k^{\text{tot}}$ falls by $f\tilde{d}_i/(S_0 - D_k)$, while the allocation excludes it; before T_k, nobody includes it, and after T_{k+1} everybody does. For the variance: the contributors form two groups, with weights f and $1 - f$, whose average $\ln F_j$ differ by $\tilde{d}_i/(S_0 - D_k)$. (iv) $D_k^{\text{tot}} - D_k = \sum_i \tilde{d}_i(\mathbf{1}_{\{t_i+\Delta\le T_k\}} - \mathbf{1}_{\{t_i\le T_k\}})$, which equals $-\tilde{d}_i$ exactly when $t_i \le T_k < t_i + \Delta$. (v) Immediate. (vi) $\delta D_k^{\text{tot}} = \tau D_k$; for a yield, $D_k = S_0(1-e^{-yT_k}) \approx S_0\, y\, T_k$ and $S_0 - D_k \approx S_0$. □

Table 1: Signatures of the mechanisms. The last column gives the typical shape tested by the detector of Section 5.

Mechanism	Carry difference δ	Repo difference	Typical shape
Panel gap (3.1)	smooth noise, increments at ex-dates	roughly flat noise	noise
Contributor drop-off (3.4)	step at the drop-off	jump decaying like $1/T$	step
(i) Spot mismatch	constant	hyperbola $-\varepsilon/T$	level
(ii) Error on one period	step after the ex-date	jump decaying like $1/T$	step
(iii) Ex-date disagreement	one pillar	spike	spike
(iv) Stale profile	pillars near the ex-dates	yearly zigzag	season
(v) Forward outlier	one pillar	spike	spike
(vi) Proportional bias	staircase; $\approx -\tau y T$ for a stream	steps; a level for an index	step
FO mismark	$-\Delta q(T)\,T$, smooth	smooth level or slope	none

Running example

Figure 4 shows the running example's stock under each of the eight benchmark scenarios, with the same contributor noise as in the clean market. In (a), the option contributors' spot is off by +0.059%: the Totem curve dives like $1/T$ at the short end. In (b), the Totem 2027 level is off by −0.40: a jump at nine months that decays like $1/T$. In (c), 38% of the option panel put the dividend going ex 19 days after the six-month expiry before it: a spike at six months. In (d), the allocation profile is shifted by −27 days: the May dividends move before the six- and eighteen-month expiries, hence two negative spikes. In (e), the consensus $\ln F$ at two years is off by +95 bp. In (f), the option contributors left after the three-year drop-off overstate later dividends by 9%: the distortion is smooth and stays inside the band. In (g), it is the FO repo that is off by +16 bp, Totem being intact. In (h), every Totem period is biased by −11% (gross/net bias).

3.4 What the data cannot tell

Theorem 3.6 (Non-identifiability of the split)**.** *Assume European option prices depend on a contributor's views only through its forwards. For every contributor j of the option panel, replace (q_j, D_j) by $(q_j + u_j, D'_j)$ with $u_j(T)\,T = \ln(S_0 - D'_j(T)) - \ln(S_0 - D_j(T))$. Then every option price is unchanged, hence every statistic of the option panel (consensus, dispersion, contributor count); the statistics of the dividend panel are unchanged too. The implied repo q^{tot} is therefore unchanged, while the option panel's average repo $\bar{q}^{\text{opt}}$ moves by the average of the u_j. Consequently, the panel gap $q^{\text{tot}} - \bar{q}^{\text{opt}}$ is not a function of the observable data.*

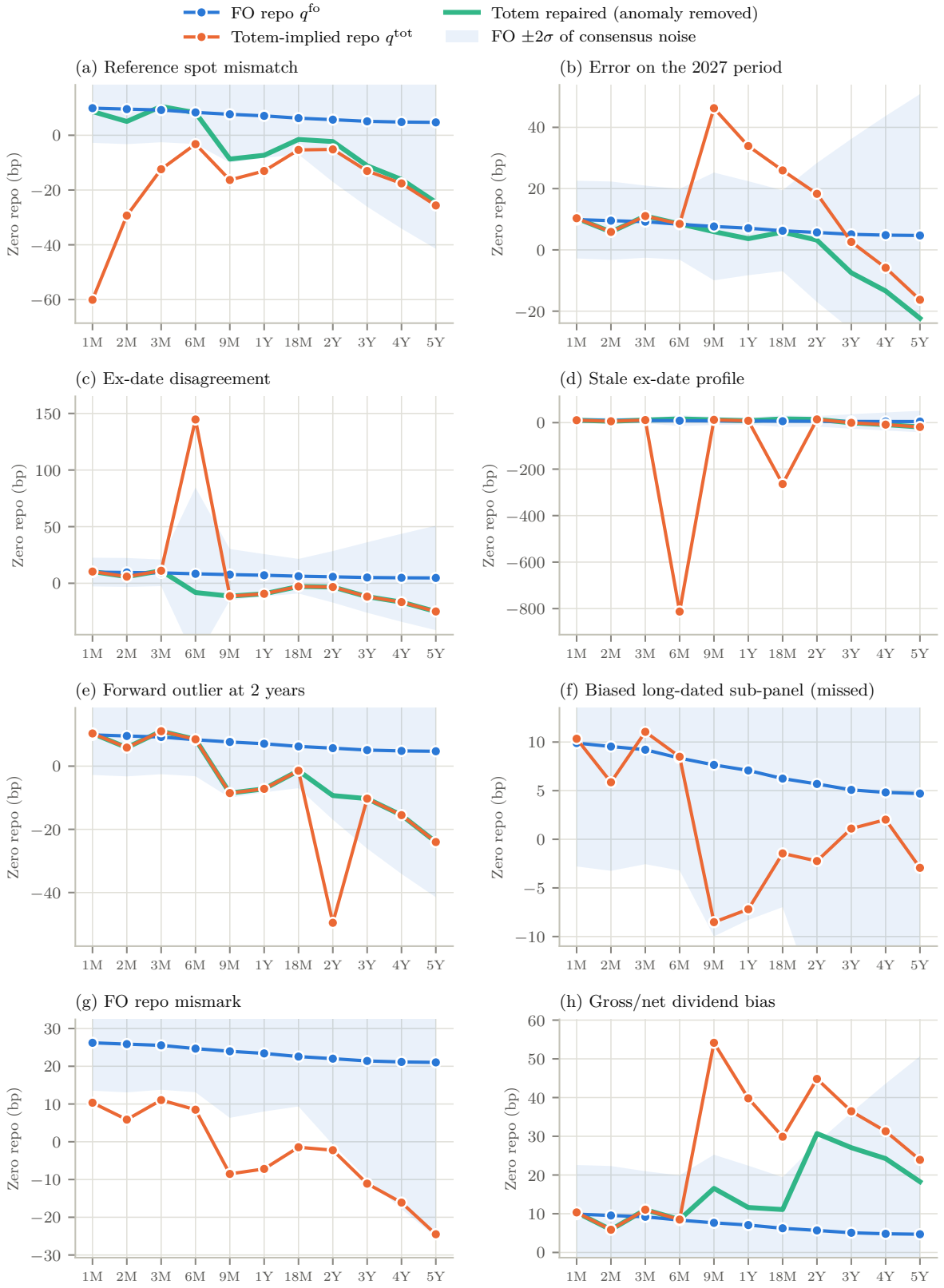


Figure 4: Running example under the eight scenarios: FO repo, Totem-implied repo, band of ±2 standard deviations of consensus noise around the FO, and, in green, the Totem curve repaired by the detector of Section 5 (anomaly removed). In (f), the detector flags nothing; in (g), it flags no Totem anomaly but a possible FO mismark. Table 3 details the diagnoses.

Proof. The transformation leaves $\ln F_j = (r - q_j)T + \ln(S_0 - D_j)$ unchanged for every j and every T: each contributor's option prices, and every statistic derived from them, are unchanged. The dividend panel is untouched. q^{tot} is computed from these statistics only, so it does not move, while $\bar{q}^{\text{opt}}$ moves by the average of the u_j, which can take any value. □

Corollary 3.7. *No estimator built from Totem consensus data alone (curve smoothing, clustering, neural network) can remove the panel gap. One can, however, compute its size (Section 5), recognise the structured mechanisms by their signature (Proposition 3.5), and do without the split between repo and dividends where the exposure does not need it (Section 4).*

Remark 3.8 (What resolves the indeterminacy). Information outside the consensus pins down one of the two sides directly: listed dividend futures, total return futures, which quote a funding spread, and stock borrow fees for the short end. Contributors common to both services shrink the gap (Corollary 3.2), and American options carry a little information on the split through early exercise.

Key points

- Implied repo = option panel repo + panel gap. The panel gap is normal noise of computable size: a few basis points at one year, one to several tens at five years.
- Real anomalies have recognisable shapes in carry: level (spot), step (dividend period, contributor drop-off), spike (ex-date, forward outlier), season (stale profile).
- The option panel's split between repo and dividends cannot be identified from Totem: no statistical method, however sophisticated, will recover it.

4 Controlling the repo from the forward and dividend controls

Goal of the section. (G2) Answer the question "I have the forward control, I have the dividend control: how much is the repo control?", and measure, book by book, what Totem's split between repo and dividends changes in the control.

4.1 Control lines

Collect the repo rates at the pillars and the dividend levels per period in $\theta = (q, \ell) \in \mathbb{R}^N \times \mathbb{R}^m$. The spot, the rate curve and the allocation profile are fixed, so the forwards at the pillars form a smooth function $F(\theta) \in \mathbb{R}^N$. By Proposition 2.10, for given forwards F and levels ℓ there is a unique repo $q = \psi(F, \ell)$ such that $F(\psi(F, \ell), \ell) = F$: this is the *implied repo map*. The FO marks are $\theta^{\text{fo}} = (q^{\text{fo}}, \ell^{\text{fo}})$. Totem provides forwards F^{tot} and levels ℓ^{tot}, hence $q^{\text{tot}} = \psi(F^{\text{tot}}, \ell^{\text{tot}})$ and $\hat{q} = \psi(F^{\text{tot}}, \ell^{\text{fo}})$. A book has a value $V(\theta)$ and FO sensitivities $s^q = \partial_q V(\theta^{\text{fo}}) \in \mathbb{R}^N$ and $s^\ell = \partial_\ell V(\theta^{\text{fo}}) \in \mathbb{R}^m$. Finally, let

$$\Delta\ell = \ell^{\text{tot}} - \ell^{\text{fo}}, \qquad \Delta q = q^{\text{tot}} - q^{\text{fo}}, \qquad \Delta\hat{q} = \hat{q} - q^{\text{fo}}.$$

Definition 4.1 (Control lines)**.**

$$\text{dividend line} = \langle s^\ell, \Delta\ell \rangle, \qquad \text{repo line} = \langle s^q, \Delta q \rangle, \qquad \text{forward line} = \langle s^q, \Delta\hat{q} \rangle.$$

The *total* is the sum of the dividend and repo lines. The forward line is the repo line one would get by keeping the FO dividends: it depends on Totem only through its forwards.

Definition 4.2 (Forward-linear book)**.** A book is *forward-linear* if $V(\theta) = \varphi(F(\theta))$ for a C^2 function φ: its value depends on the repo and the dividends only through the forwards. This

is the case for European options valued with a volatility surface that does not depend on the split, for forwards and for futures.

4.2 The Jacobian of the implied repo

Lemma 4.3 (Jacobian)**.** *The matrix $J = \partial_\ell \psi(F, \ell) \in \mathcal{M}_{N,m}(\mathbb{R})$ is $J = -(\partial_q F)^{-1} \partial_\ell F$. The matrix $\partial_q F$ is lower triangular with diagonal terms $\partial F_k / \partial q_k \le -T_k F_k < 0$, hence invertible. To first order,*

$$J_{kp} \approx -\frac{a_{kp}}{T_k\,(S_0 - D_k)}. \tag{8}$$

Proof. Differentiate $F\big(\psi(F,\ell),\ell\big) = F$ with respect to ℓ: $\partial_q F\, J + \partial_\ell F = 0$. With a constant forward repo between pillars, F_k depends only on $q_1, \dots, q_k$, hence the triangular form; the diagonal term is $T_k\, \mathcal{F}_k'(q_k T_k) \le -T_k F_k$ by Proposition 2.10. For the first order, hold the forwards fixed in (2): $T_k\, \delta q_k = -\delta D_k/(S_0 - D_k) = -a_{kp}\, \delta \ell_p/(S_0 - D_k)$. □

4.3 The control identity

Theorem 4.4 (Control identity)**.** *For every book,*

$$\textit{dividend line} + \textit{repo line} = \textit{forward line} + \langle \eta, \Delta\ell \rangle + O(\|\Delta\|^2), \tag{9}$$

where $\eta = s^\ell + J^\top s^q \in \mathbb{R}^m$, J is evaluated at $(F^{\mathrm{tot}}, \ell^{\mathrm{fo}})$ and $\|\Delta\| = \|\Delta\ell\| + \|\Delta\hat{q}\|$. We call $\langle \eta, \Delta\ell \rangle$ the non-forward line *and η the* non-forward exposure*. If the book is forward-linear, $\eta = O(\|\Delta\hat{q}\|)$: the total equals the forward line up to second order, and therefore does not depend on the Totem dividends to first order.*

Proof. Expanding ψ in ℓ gives $\psi(F^{\mathrm{tot}}, \ell^{\mathrm{tot}}) = \psi(F^{\mathrm{tot}}, \ell^{\mathrm{fo}}) + J\Delta\ell + O(\|\Delta\ell\|^2)$, that is, $\Delta q = \Delta\hat{q} + J\Delta\ell + O(\|\Delta\ell\|^2)$. Hence

$$\langle s^\ell, \Delta\ell \rangle + \langle s^q, \Delta q \rangle = \langle s^q, \Delta\hat{q} \rangle + \langle s^\ell + J^\top s^q, \Delta\ell \rangle + O(\|\Delta\|^2).$$

If $V = \varphi \circ F$, then ${s^q}^\top = \nabla\varphi\, \partial_q F(\theta^{\mathrm{fo}})$ and ${s^\ell}^\top = \nabla\varphi\, \partial_\ell F(\theta^{\mathrm{fo}})$, so $\eta^\top = \nabla\varphi\, [\partial_\ell F + \partial_q F\, J](\theta^{\mathrm{fo}})$. At the point $\theta' = (\hat{q}, \ell^{\mathrm{fo}})$, the bracket vanishes by Lemma 4.3; since $\theta' - \theta^{\mathrm{fo}} = (\Delta\hat{q}, 0)$ and the derivatives are Lipschitz, the bracket at θ^{fo} is $O(\|\Delta\hat{q}\|)$, and $\langle \eta, \Delta\ell \rangle = O(\|\Delta\|^2)$. □

Answer to the controller's question

Rewriting (9):

$$\boxed{\text{repo line} = \text{forward line} - \text{dividend line} + \text{non-forward line}}$$

up to second order. The repo control follows from the forward control and the dividend control. For a forward-linear book, the non-forward line is zero: the repo control is exactly the difference of the other two, and the total (dividend + repo) does not depend on Totem's dividend contributors. The dividend panel enters only through the non-forward exposure $\eta = s^\ell + J^\top s^q$, which is computed from the FO sensitivities and the bootstrap Jacobian alone, with no revaluation.

The result even holds exactly under full revaluation.

Theorem 4.5 (Exactness under full revaluation)**.** *If the book is forward-linear and all its maturities are Totem pillars, the full-revaluation control $V(\theta^{\mathrm{tot}}) - V(\theta^{\mathrm{fo}}) = \varphi(F^{\mathrm{tot}}) - \varphi(F(\theta^{\mathrm{fo}}))$, with $\theta^{\mathrm{tot}} = (q^{\mathrm{tot}}, \ell^{\mathrm{tot}})$, does not depend on the Totem dividends ℓ^{tot}.*

Proof. By construction of the implied repo, $F(\theta^{\text{tot}}) = F(\psi(F^{\text{tot}}, \ell^{\text{tot}}), \ell^{\text{tot}}) = F^{\text{tot}}$, whatever ℓ^{tot}. □

Remark 4.6 (Maturities between two pillars). For a maturity between two pillars, the forward depends on the interpolated carry and on the dividends going ex in between: a residual dependence on ℓ^{tot} remains. It is of the order of the dividends going ex between the maturity and the neighbouring pillars, that is, exactly the ex-date and profile effects of Proposition 3.5, which the detector watches. The second-order term of (9) is written out in Appendix A.5.

4.4 Examples

Example 4.7 (A call)**.** A one-year at-the-money call: spot 100, rate 3%, volatility 25%, FO dividend $D^{\text{fo}} = 3.5$, FO repo 60 bp. The Totem forward is 20 bp below the FO one, and we try three dividend panels:

D^{tot}	q^{tot} (bp)	Dividend line	Repo line	Total	Full revaluation
3.5	80	0.000	−0.102	−0.102	−0.102
3.0	132	0.264	−0.365	−0.101	−0.102
4.2	7	−0.370	0.269	−0.101	−0.102

The implied repo moves by 125 bp from one panel to the next and each line by about 0.3, but the total does not move, and it equals the full revaluation.

Example 4.8 (Three books)**.** The prototype's pricer values three books on eleven repo pillars and six dividend periods: A (calls, puts and a forward), B (A plus a short 2028 dividend swap) and C (A plus a one-year stock loan fee). The Totem forwards are 20 bp below the FO ones, and the Totem dividend panel shifts the 2027 and 2028 periods by $\Delta\ell \in \{-0.4, 0, 0.5\}$:

	$\Delta\ell$	Dividend line	Repo line	Total	Forward line	Non-forward line	Full revaluation
A	−0.4	5.16	−6.91	−1.75	−1.74	−0.01	−1.74
	0.0	0.00	−1.74	−1.74	−1.74	0.00	−1.74
	+0.5	−6.46	4.74	−1.72	−1.74	0.01	−1.74
B	−0.4	8.15	−6.91	1.24	−1.74	2.98	1.25
	0.0	0.00	−1.74	−1.74	−1.74	0.00	−1.74
	+0.5	−10.19	4.74	−5.45	−1.74	−3.72	−5.47
C	−0.4	5.16	−5.16	0.00	−1.18	1.17	0.01
	0.0	0.00	−1.18	−1.18	−1.18	0.00	−1.18
	+0.5	−6.46	3.82	−2.64	−1.18	−1.47	−2.66

For A, the total stays at −1.74 (the forward line) while each line moves by more than six units. For B and C, the total moves, by exactly the non-forward line: the dividend swap is a genuine dividend exposure, the stock loan fee a genuine repo exposure. Full revaluation confirms every line (Fig. 5).

4.5 What the split costs a book

Proposition 4.9 (Materiality of the split)**.** *Assume the consensus error on the dividend levels $\Delta\ell$ has covariance matrix Γ^{ℓ}, the Totem forwards being fixed. To first order, the variance of*

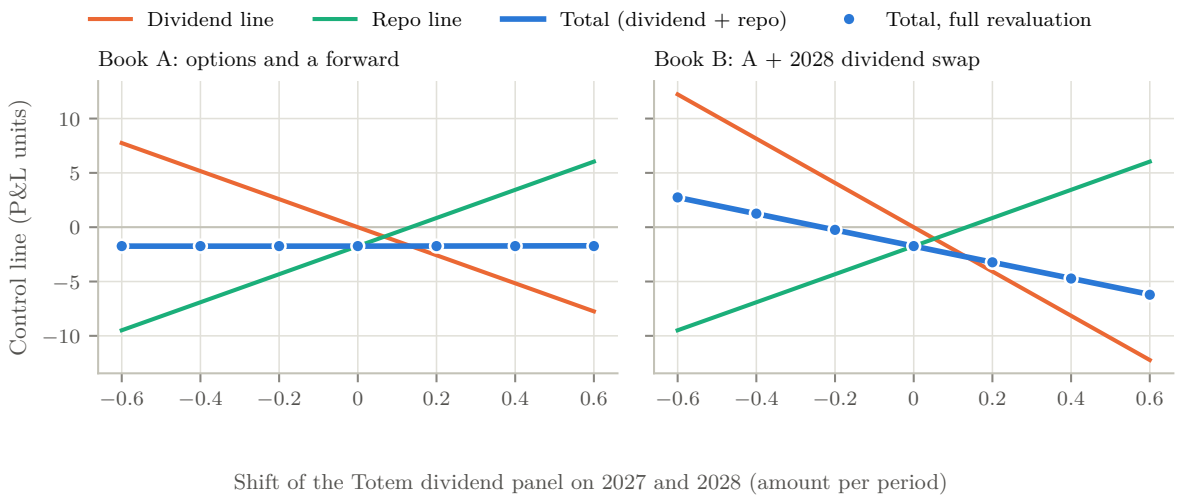


Figure 5: Dividend and repo lines, and their total, as the Totem dividend panel moves (book A on the left, B on the right). Dots: full revaluation. For a forward-linear book, the total is flat (Theorems 4.4 and 4.5); the dividend swap adds a genuine dividend exposure, whose line is exactly $\langle \eta, \Delta\ell \rangle$.

the total is $\eta^\top \Gamma^\ell \eta$*, against* $(s^\ell)^\top \Gamma^\ell s^\ell$ *for the dividend line alone. The share of the dividend line's uncertainty cancelled by the repo line is therefore*

$$1 - \sqrt{\frac{\eta^\top \Gamma^\ell \eta}{(s^\ell)^\top \Gamma^\ell s^\ell}}.$$

Proof. By Theorem 4.4, the total is the forward line, which does not depend on ℓ^{tot}, plus $\langle \eta, \Delta\ell \rangle$. □

Example 4.10 (Materiality)**.** With a consensus dispersion of 5% of each period, ten contributors and a correlation $e^{-|c_p - c_{p'}|/2}$ between periods with mid-points c_p and $c_{p'}$:

Book	Std. dev. of the dividend line	Std. dev. of the total	Share cancelled
A	0.725	0.001	99.8%
B	1.113	0.436	60.8%
C	0.725	0.162	77.6%

99.8% of the dividend line's uncertainty cancels for book A, 60.8% for B and 77.6% for C.

4.6 When the split matters

The non-forward exposure η is not zero for dividend swaps and futures (pure dividend exposure), for stock borrow fees and the funding legs of total return swaps (pure repo exposure), for American exercise (the early-exercise boundary depends on dividends and repo separately) and for exotics valued with discrete dividend dynamics. For these books, the split must be controlled directly, and the quality of the implied repo curve matters: this is the subject of Section 5.

Key points

- Control the *forward line* (Totem forwards, FO dividends) and the *non-forward line* $\langle \eta, \Delta\ell \rangle$; the latter needs the Totem dividends only where $\eta \neq 0$.
- Large dividend and repo lines of opposite signs, with a small total, are a split artefact, not a mismark.
- In prudent valuation [4], aggregating uncertainties per parameter without offsetting counts a split artefact twice; taking the forward as the parameter where $\eta \approx 0$ removes the double count.

5 Detecting bad curves

Goal of the section. (G3) Build a test that flags a bad Totem curve with a known false alarm rate, names the mechanism at work, gives its size in natural units, repairs the curve, and tells a Totem anomaly from an FO mismark.

5.1 The principle in five steps

1. Compute the carry difference δ and the covariance of its noise from the dispersions and contributor counts that Totem publishes (Section 5.2).
2. Remove a smooth repo difference, $\beta_0 + \beta_1 \ln T$: it is a legitimate disagreement between Totem and the FO, not a curve anomaly (Section 5.3).
3. Test the typical shapes of Proposition 3.5 one by one with a matched filter, with a common threshold that fixes the false alarm rate of the whole family of tests (Section 5.4).
4. On an alarm, select the shapes that explain the difference, give their sizes in natural units and the repaired curve (Section 5.5).
5. Independently, test the level β_0 to flag a possible FO mismark (Section 5.5).

5.2 The noise model

Definition 5.1 (Observation model)**.** The detector models the carry difference as

$$\delta = X\beta + \sum_{g \in \mathcal{S}} a_g \, g + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \kappa\Gamma), \tag{10}$$

where $X \in \mathcal{M}_{N,2}(\mathbb{R})$ has columns $(T_k)_k$ and $(T_k \ln T_k)_k$, so that $X\beta$ is the smooth repo difference $\beta_0 + \beta_1 \ln T$ written in carry; $\mathcal{S}$ is a possibly empty set of typical shapes $g \in \mathbb{R}^N$ with sizes a_g; Γ is the covariance of the consensus noise, built below from Totem's statistics; and $\kappa > 0$ is a single scale, calibrated on clean history. Under the null hypothesis, $\mathcal{S} = \emptyset$.

By Lemma 2.2 and Proposition 3.1, the consensus noise in carry is $\varepsilon_k = -e_k^{\text{opt}} - e_k^{\text{div}}$, where e_k^{opt} is the error of the option panel's consensus $\ln F_k^{\text{tot}}$ and $e_k^{\text{div}} = \delta D_k^{\text{tot}}/(S_0 - D_k)$ that of the allocated consensus dividends. We take the two panels as independent ($n^{\text{com}} = 0$, the cautious case), so $\Gamma = \Gamma^{\text{opt}} + \Gamma^{\text{div}}$. For each maturity, Totem publishes the dispersion σ_k^F of the contributors' forwards, hence the relative dispersion $s_k = \sigma_k^F / F_k^{\text{tot}}$, and, for each period, the dispersion σ_p^ℓ of the dividend levels.

Proposition 5.2 (Noise covariance)**.** *Split the relative forward dispersion into a repo part and a dividend part, $s_k^2 = \sigma_q^2 T_k^2 + s_k'^2$, where σ_q is the dispersion of the contributors' repo*

views. Assume the repo views persist across maturities, and the dividend views are made of a persistent factor, with weight $w \in [0,1]$, plus independent increments. Then

$$\Gamma^{\text{opt}}_{kl} = \frac{\sigma_q^2\, T_k T_l + w\, s'_k s'_l + (1-w)\min(s'_k, s'_l)^2}{\max(n_k^{\text{opt}}, n_l^{\text{opt}})}, \tag{11}$$

$$\Gamma^{\text{div}} = A_S\, \Gamma^{\ell}\, A_S{}^{\top}, \qquad (A_S)_{kp} = \frac{a_{kp}}{S_0 - D_k}, \qquad \Gamma^{\ell}_{pp'} = \frac{\rho_{pp'}\, \sigma^{\ell}_p \sigma^{\ell}_{p'}}{\max(n_p^{\text{div}}, n_{p'}^{\text{div}})}, \tag{12}$$

where $\rho_{pp'}$ is the correlation of the view errors between periods p and p'.

The proof is in Appendix A.3. The prototype takes $w = 1/2$, $\rho_{pp'} = e^{-|c_p - c_{p'}|/2}$ for periods with mid-points c_p and $c_{p'}$ (in years), and reads σ_q off the short end, where dividend uncertainty is negligible: it is the median of s_k/T_k over the first three pillars (the minimum, biased low, would understate the uncertainty on the repo level). A diagonal term accounts for the residual of the parity fit. The scale κ absorbs a common bias in these settings; it is the *shape* of Γ that makes the detector powerful, because it tells it which differences are cheap (smooth, long-dated) and which are expensive (short-dated, localised).

Dispersion spikes are symptoms. When part of the contributors put a dividend on the other side of an expiry (Proposition 3.5, (iii)), the published dispersion at that expiry rises. Keeping it as is in Γ would hide the very anomaly it signals.

Proposition 5.3 (Masking)**.** *In situation (iii) of Proposition 3.5, let s_k be the dispersion without the disagreement and $\Delta = \tilde{d}_i/(S_0 - D_k)$. The statistic of the shift at T_k, computed with the published dispersion, is*

$$z = \frac{f\,\Delta}{\sqrt{s_k^2 + f(1-f)\Delta^2}/\sqrt{n_k^{\text{opt}}}} \;\le\; \sqrt{\frac{n_k^{\text{opt}}\, f}{1-f}}$$

whatever the size of the dividend, while with the dispersion without the disagreement, z grows linearly in Δ.

Proof. The mean shift and the extra variance are given by Proposition 3.5, (iii); as $\Delta \to +\infty$, the ratio increases to $\sqrt{n_k^{\text{opt}}}\, f/\sqrt{f(1-f)}$. □

With $f = 0.375$ and $n_k^{\text{opt}} = 16$, the bound is 3.1: barely the test threshold, so the disagreement would often go unnoticed. The prototype therefore caps an isolated dispersion spike at 1.5 times the larger of the two neighbouring dispersions before building Γ, and publishes the ratio to the neighbour, which is itself an indicator of disagreement. In the running example, this ratio is 16 at six months in the ex-date disagreement scenario, and stays below 2 everywhere else.

5.3 The null model: what is not an anomaly

The term $X\beta$ represents a smooth repo difference between Totem and the FO: β_0 is the repo difference at one year (since $\ln 1 = 0$) and β_1 its slope per unit of $\ln T$. Such a difference can come from a genuine difference of view on the repo, from an FO mismark or from a smooth bias in the consensus; in every case, it is not a *bad* curve. The detector removes it from every test and publishes it separately: $\hat{\beta}_0$ feeds the FO mismark test (Section 5.5). If two parameters turn out too rigid on real data, a penalised spline can replace $X\beta$ [1]; nothing else changes.

5.4 Typical shapes and the test

Each mechanism of Proposition 3.5 defines a typical shape $g \in \mathbb{R}^N$, tested at every possible position (Table 2). With eleven pillars, this gives 22 typical shapes.

Table 2: The detector's typical shapes. $\mathbf{1}$ is the vector with all components equal to 1, e_k the k-th canonical basis vector and K_h a Gaussian kernel of width $h = 20$ days. The natural units follow from Proposition 3.5. A spike at the last pillar is the same vector as a step at that pillar.

Typical shape	Vector g	Mechanisms	Size a in natural units
level	$\mathbf{1}$	spot mismatch; dividend before T_1	spot: $\varepsilon \approx -a$; dividend: $\delta d \approx -a\,(S_0 - D_1)$
step at T_k	$\left(\mathbf{1}_{\{l \geq k\}}\right)_{l \in \{1,\dots,N\}}$	dividends going ex in $(T_{k-1}, T_k]$; contributor drop-off at T_k	$\delta\ell \approx -a\,(S_0 - D_k)$
spike at T_k	e_k	forward outlier; ex-date disagreement around T_k	$\mu \approx -a$; share of the dividend $f \approx a\,(S_0 - D_k)/\tilde{d}_i$
season	$\left(\sum_i \dfrac{\tilde{d}_i\, K_h(T_k - t_i)}{S_0 - D_k}\right)_k$	shifted ex-date profile	shift $\Delta \approx a$ years

Whitening and projection. Let $W \in \mathcal{M}_N(\mathbb{R})$ be a whitening matrix, with $W(\kappa\Gamma)W^\top = I_N$ (for example $W = L^{-1}$ with the Cholesky factorisation $\kappa\Gamma = L\,L^\top$), and Π the orthogonal projection onto the orthogonal complement of the columns of WX. Let

$$\hat{\varepsilon} = \Pi W \delta \quad \text{(whitened residual)}, \qquad \tilde{g} = \Pi W g \quad \text{(whitened typical shape)}.$$

After whitening, the noise is a Gaussian vector with identity covariance, and the projection removes the smooth repo difference.

Theorem 5.4 (Likelihood ratio test for one typical shape)**.** *In the model $\delta = X\beta + a\,g + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \kappa\Gamma)$, with β unknown, the generalised likelihood ratio statistic of $H_0 : a = 0$ against $H_1 : a \neq 0$ is z_g^2, with*

$$z_g = \frac{\langle \tilde{g}, \hat{\varepsilon} \rangle}{\|\tilde{g}\|} \sim \mathcal{N}(a\|\tilde{g}\|, 1). \tag{13}$$

The maximum likelihood estimate of the size is $\hat{a}_g = \langle \tilde{g}, \hat{\varepsilon} \rangle / \|\tilde{g}\|^2$, with standard deviation $1/\|\tilde{g}\|$. Among tests invariant under the null subspace and the sign of a, the one that rejects for large values of $|z_g|$ is uniformly most powerful.

Proof. We have $W\delta = WX\beta + a\,Wg + W\varepsilon$ with $W\varepsilon \sim \mathcal{N}(0, I_N)$; the maximised log-likelihood is, up to a constant, $-\frac{1}{2}$ times the residual sum of squares. Under H_0, this sum is $\|\hat{\varepsilon}\|^2$; under H_1, by the Frisch–Waugh–Lovell theorem [2], it is $\|\hat{\varepsilon}\|^2 - \langle \tilde{g}, \hat{\varepsilon} \rangle^2 / \|\tilde{g}\|^2$. Hence $2 \ln \Lambda = z_g^2$, and $\hat{a}_g$ is the partial regression coefficient. Since $\Pi WX = 0$ and Π is symmetric and idempotent, $\hat{\varepsilon} = a\,\tilde{g} + \Pi W\varepsilon$ and $\langle \tilde{g}, \hat{\varepsilon} \rangle = a\|\tilde{g}\|^2 + \langle \tilde{g}, W\varepsilon \rangle$, where $\langle \tilde{g}, W\varepsilon \rangle \sim \mathcal{N}(0, \|\tilde{g}\|^2)$. Optimality is the result on matched subspace detectors of Scharf and Friedlander [5]. □

Corollary 5.5 (Power)**.** *At threshold c, an anomaly of shape g is detected with probability $\Phi(\lambda - c) + \Phi(-\lambda - c)$, where Φ is the standard normal distribution function and $\lambda = |a|\,\|\tilde{g}\|$ the* signal-to-noise ratio *of the anomaly. At the benchmark's typical threshold, $c \approx 3.39$, an anomaly with $\lambda = 4$ is detected with probability 73 % and one with $\lambda = 6$ with probability 99.5 %.*

Proposition 5.6 (Global test and common threshold)**.** *Under H_0:*

(i) $\|\hat{\varepsilon}\|^2$ *follows a* χ^2 *distribution with* $N-2$ *degrees of freedom; this global test catches a roughness that no typical shape explains;*

(ii) the vector $(z_g)_g$ *is centred Gaussian with correlation matrix* $\Omega_{gh} = \langle \tilde{g}, \tilde{h} \rangle / (\|\tilde{g}\| \|\tilde{h}\|)$*, which depends neither on* β *nor on* κ*; the threshold* c_α *such that* $\mathbb{P}(\max_g |z_g| > c_\alpha) = \alpha/2$ *is computed exactly by simulating* $\mathcal{N}(0, \Omega)$*, and it is below the Šidák threshold [6];*

(iii) the rule "alarm if $\max_g |z_g| > c_\alpha$ *or if* $\|\hat{\varepsilon}\|^2$ *exceeds its* $1-\alpha/2$ *quantile" has a false alarm rate of at most* α*.*

Proof. (i) $\hat{\varepsilon} = \Pi W \varepsilon$ and Π is an orthogonal projection of rank $N-2$. (ii) z_g is a linear form of $W\varepsilon$; Šidák's inequality gives $\mathbb{P}(\max_g |z_g| \le c) \ge \prod_g \mathbb{P}(|z_g| \le c)$ for every correlation. (iii) Union bound. □

Proposition 5.7 (Exact calibration of the scale)**.** *Replacing* $\kappa = 1$ *by* κ *divides every* z_g *by* $\sqrt{\kappa}$ *and* $\|\hat{\varepsilon}\|^2$ *by* κ*, without changing* c_α*. On clean history, the alarm rate is therefore a decreasing function of* κ*, and the smallest* κ *whose alarm rate does not exceed the target* α *is found exactly by bisection.*

Proof. Whitening with $\kappa\Gamma$ multiplies W by $\kappa^{-1/2}$; Π and Ω do not change. □

5.5 Explanation, repaired curve and FO mismark

Explanation. After an alarm, typical shapes are selected one at a time: at each step, the one with the largest $|z|$ after projecting out the null model and the shapes already selected is added, as long as it exceeds c_α (matching pursuit [3]). The selected shapes are then refitted jointly with the null model by generalised least squares, and each size is translated into natural units (Table 2).

Repaired curve. The repaired Totem curve removes the anomaly and keeps everything else:

$$q_k^{\text{rep}} = q_k^{\text{tot}} - \frac{1}{T_k} \sum_{g \in \mathcal{S}} \hat{a}_g \, g_k.$$

Shapes contained in the null subspace are undetectable by construction: a smooth repo difference is a legitimate disagreement, published but never flagged as a curve anomaly.

FO mismark. The level $\hat{\beta}_0$ (Totem minus FO, at one year) and its standard deviation come from the same fit. The FO repo is flagged as a possible mismark when $|\hat{\beta}_0|$ exceeds both 10 bp and $t_{\nu,1-\alpha/2}\, s\, \sigma(\hat{\beta}_0)$, where:

- $t_{\nu,1-\alpha/2}$ is a quantile of the Student distribution with $\nu = n^{\text{opt}} - 1$ degrees of freedom at the short end: the standard deviation rests on dispersions estimated from a handful of contributors (Appendix A.4);
- $s \ge 1$ is a scale calibrated on clean history like κ, so that clean names raise at most a proportion α of mismark flags; it is 1.00, 1.05, 1.00 and 1.18 for the benchmark's four categories.

The full algorithms (bootstrap, detection, calibration) are in Appendix B.

5.6 A worked example

Example 5.8 (Four pillars)**.** Pillars $T = (0.25, 0.5, 1, 2)$, independent noise with standard deviations $(1, 2, 4, 8)$ bp of carry, a null model reduced to the repo level ($X = T$), a true level of 30 bp,

a spike of 12 bp at T_2 and noise $(0.4, -1.1, 2.6, -3.0)$ bp, hence $\delta = (7.9, 25.9, 32.6, 57.0)$ bp. The fitted level is $\hat{\beta}_0 = 36.1$ bp and $\|\hat{\varepsilon}\|^2 = 21.0$ (the 99% quantile of $\chi^2(3)$ is 11.3). The 7 typical shapes give:

Typical shape g	z_g	$\hat{a}_g$ (bp)	Std. error (bp)
Level	0.55	0.8	1.5
Step from T_2	4.06	11.0	2.7
Step from T_3	−2.21	−10.7	4.8
Step at T_4	−2.20	−20.3	9.2
Spike at T_1	−1.31	−1.5	1.2
Spike at T_2	4.52	10.4	2.3
Spike at T_3	−1.02	−4.7	4.6

The common threshold at $\alpha = 2\%$ is $c_\alpha = 3.11$ (Šidák: 3.19). The spike at T_2 wins ($z = 4.52$, size 10.4 ± 2.3 bp for 12 injected); the step from T_2 also scores high, because the two shapes are correlated. Matching pursuit selects the spike; after that, all remaining $|z|$ are below 0.7 and $\|\hat{\varepsilon}\|^2$ falls to 0.6: one explanation is enough.

5.7 Back to the running example

Running example

Table 3 gives, for the running example's stock, the detector's diagnosis in the nine situations of Fig. 4, with the scale $\kappa = 1.60$ calibrated for European stocks and $\alpha = 2\%$. The clean market triggers nothing ($\max_g |z_g| = 2.26 < c_\alpha = 3.42$, and the global test has a p-value of 0.42).

Several points stand out.

- **The mechanisms are named correctly.** Spot, dividend period, ex-date, profile and forward outlier are all attributed to the right shape and the right maturity. The spot mismatch is recovered at +0.057% for +0.059% injected; the forward outlier at +81 bp (standard deviation 12) for +95 bp.
- **Sizes are estimates.** The 2027 error is estimated at −0.29 (standard deviation 0.05) for −0.40 injected: the clean market's own noise at nine months (the panel gap of Fig. 2) is netted from the estimate. The share of the misplaced dividend is estimated at 18% for 38% injected, because the consensus outlier filter has already removed part of the disagreeing contributors. The profile shift is estimated at −46 days for −27: the conversion into days is a linearisation, right in sign and order of magnitude.
- **What is smooth is not flagged.** The biased long-dated sub-panel produces a smooth distortion after the 3Y drop-off; it is absorbed by the smooth repo difference and the detector flags nothing ($\max_g |z_g| = 2.31$). This is Theorem 3.6 at work. Likewise, the gross/net bias is seen as a step at the first ex-date (−0.27 on the 2027 dividend) plus a repo slope, as Proposition 3.5, (vi) predicts.
- **An FO error is not a bad curve.** When the FO repo is wrong, no curve anomaly is flagged, and the level test finds $\hat{\beta}_0 = -17.8$ bp (standard deviation 5.4), beyond the 14.1 bp threshold: possible FO mismark.

Table 3: Running example: what is injected and what the detector says. z is statistic (13) of the first selected shape; sizes are translated into natural units (Table 2).

Situation	What is injected	What the detector says
Clean	no anomaly	no alarm ($\max_g \lvert z_g\rvert = 2.26 < c_\alpha = 3.42$)
Spot mismatch	option contributors price off a spot shifted by +0.059%	level, $z = -7.0$: spot off by +0.06%
Error on one period	Totem 2027 level: 3.82 instead of 4.22	step at 9M, $z = 6.3$: dividend going ex in (6M, 9M] lower by 0.29 (−7%) in the dividend service than in the forwards
Ex-date disagreement	38% of the option panel put the dividend going ex 19 days after the 6M expiry before it	spike at 6M, $z = 4.1$: about 18% of the dividend on the other side of the expiry; forward dispersion 16 times its neighbours'
Stale profile	allocation profile shifted by −27 days	season, $z = -55.4$: profile shifted by about −46 days; then spike at 6M ($z = -5.4$)
Forward outlier	consensus $\ln F$ at 2Y shifted by +95 bp (5.3 standard deviations)	spike at 2Y, $z = -6.7$: $\ln F$ off by +81 bp
Biased long sub-panel	from 3Y on, the 7 remaining option contributors overstate later dividends by 9%	no alarm ($\max_g \lvert z_g\rvert = 2.31 < c_\alpha = 3.43$)
FO repo mismark	FO repo shifted by +16 bp, Totem untouched	no Totem anomaly; level $\hat{\beta}_0 = -17.8$ bp (standard error 5.4), beyond the threshold: possible FO mismark
Gross/net bias	every Totem period multiplied by 0.89	step at 9M, $z = 5.9$: dividend going ex in (6M, 9M] lower by 0.27 (−6%) in the dividend service than in the forwards

Key points

- The detector judges the carry difference against a noise computed from Totem's statistics, removes the smooth repo difference, then tests one typical shape per mechanism.
- Each test is an exact matched filter; the common threshold and the scale κ fix the false alarm rate exactly; the power is known in closed form.
- Every alarm comes with an explanation in natural units and a repaired curve; the level $\hat{\beta}_0$ separates FO mismarks from Totem anomalies.

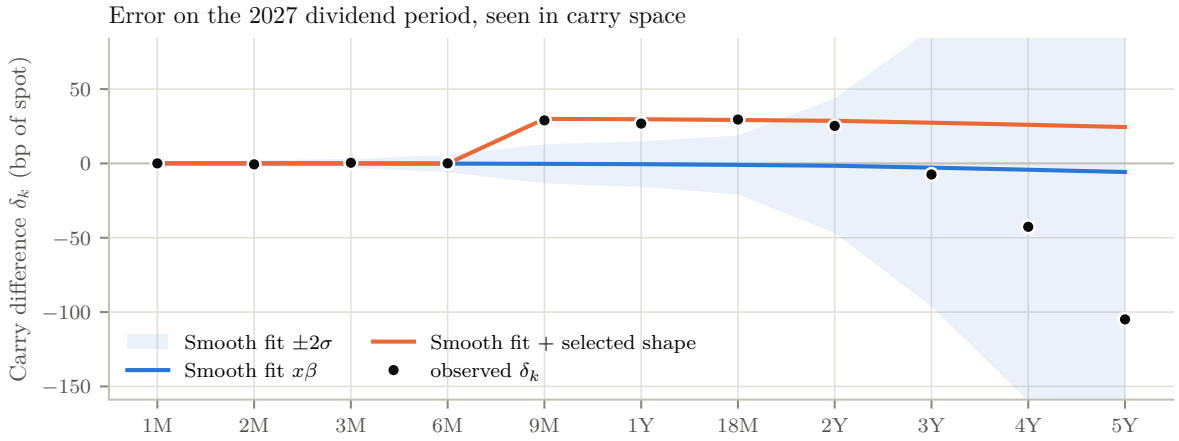


Figure 6: Running example, error on the 2027 period of Fig. 4(b), seen in carry: observed difference, fitted smooth repo difference with its band of ±2 standard deviations, and the smooth fit plus the selected step. At long maturities the noise is so large that points stray from the fit without being significant.

6 Benchmark and comparison with other approaches

> **Goal of the section.** Measure, on a synthetic Totem market built contributor by contributor, what the structural detector adds to naive checks and to the starting ideas: B-spline plus Gaussian mixture, small neural networks.

6.1 Design

The synthetic market is built from the contributors up, so that each mechanism of Section 3 appears as it does in Totem, not as a shape added to a curve. Details are in Appendix C.

- **Four categories** (Table 4), eleven maturities from one month to five years, six dividend periods (the rest of 2026, then 2027 to 2031), valuation date 14 October 2026.
- **Contributors** have consistent views (Assumption 2.5): dividends made of a persistent factor and a noise specific to each dividend, whose dispersion grows with maturity, and a smooth repo curve. Option contributors price calls and puts at five strikes with their own forward and a noisy volatility surface. Panels shrink with maturity and are nested; option and dividend panels are drawn independently.
- **The consensus** is the mean after a median/median-absolute-deviation filter; dispersions and contributor counts are published as Totem would. The controller's pipeline then runs unchanged.
- **The FO marks** are the true dividends with a 1% noise, and the true repo with a level error of 3 bp (one standard deviation).
- **The scenarios** (Table 5) are simulated for each test name with the same draw of contributors: differences between scenarios are paired, and the signal-to-noise ratio of each injection is known exactly.

Detectors. Five detectors are compared: (1) *naive gap*: the largest $|q_k^{\text{tot}} - q_k^{\text{fo}}|$; (2) *naive shape*: the largest distance of the Totem repo from the interpolation of its two neighbours; (3) *B-spline plus Gaussian mixture on the curve*: the six coefficients of a cubic B-spline fitted to the Totem curve (knots in $\ln T$), clustered by a Gaussian mixture per category whose number

Table 4: The four categories of the synthetic market.

Category	Dividends	Yield	Repo (bp), short → long	n^{opt}	n^{div}
EU stock	annual, Apr–May (35% with a Nov. interim)	2.0%–6.0%	$[0, 50] \to [0, 60]$	$16 \to 6$	$12 \to 4$
US stock	quarterly	0.5%–3.0%	$[0, 40] \to [0, 50]$	$18 \to 5$	$10 \to 3$
Hard to borrow	annual or quarterly	0.0%–2.0%	$[150, 800] \to [30, 150]$	$12 \to 4$	$8 \to 3$
EU index	monthly stream, 60% in Apr–Jun	2.8%–3.6%	$[-20, 60] \to [-10, 80]$	$25 \to 12$	$20 \to 10$

Table 5: Benchmark scenarios. The first six are anomalies of the Totem curve; for the last two, a Totem anomaly alarm is not the desired outcome (see the text).

Scenario	Injection	Size
Spot mismatch	the option contributors price off another spot	±0.03% to ±0.15%
Error on one period	one period (2027 to 2029) wrong in the dividend consensus	±0.15 to ±0.5 for a spot of 100
Ex-date disagreement	35% to 70% of the option panel put the dividend closest to an expiry on its other side	one dividend
Stale profile	allocation profile shifted so that a dividend crosses an expiry	3 to 20 days beyond
Forward outlier	consensus calls shifted at one expiry	3 to 6 standard deviations of carry noise
Biased long sub-panel	the contributors left after the largest drop-off are biased on later dividends	±8% to ±20%
FO repo mismark	FO repo shifted in parallel, Totem intact	±15 to ±50 bp
Gross/net bias	every Totem period multiplied by $1 + \tau$	$\lvert\tau\rvert$ from 8% to 18%

of components is chosen by BIC; the score is the negative log-likelihood; (4) *the same on the Totem–FO difference*; (5) *the structural detector* of Section 5.

Protocol. 300 clean names per category calibrate each detector to 2% false alarms (a score quantile for the first four; κ and the level scale for the structural detector). Then 150 new names per category are simulated under each of the 9 scenarios, that is, 5400 test names.

6.2 Results

Reading the results.

- **On average, the best methods are level.** Over the six Totem anomalies, the structural detector flags 73% of the names, the mixture on the difference 73%, on the curve 69%, the naive shape check 64% and the naive gap 57%.
- **Forward outliers** are where carry space and the noise model pay off most: 97% detected, against 35% for the naive gap, which cannot see a three- to six-sigma outlier under the long-end noise, and 87% for the naive shape check.
- **Spot mismatches** (98%) and **stale profiles** (94%) are easy for every method; the structural detector adds the explanation (Fig. 8).
- **Errors on one period** (55%, against 29% for the naive gap) are limited by their size: 0.15 to 0.5 for a spot of 100 is small against the dispersion of distant dividends, and

Table 6: Share of names flagged, each detector being calibrated to 2% false alarms on clean history. The "Clean" row is the false alarm rate realised on new names. For the last two rows before the mean, a Totem anomaly alarm is not the desired outcome.

Scenario	Naive gap	Naive shape	Spline + mixture, curve	Spline + mixture, difference	Structural detector
Clean	2%	3%	2%	3%	3%
Spot mismatch	89%	64%	96%	100%	98%
Error on one period	29%	41%	47%	56%	55%
Ex-date disagreement	55%	74%	65%	69%	71%
Stale profile	90%	92%	94%	94%	94%
Forward outlier	35%	87%	74%	78%	97%
Biased long sub-panel	42%	24%	36%	39%	24%
FO repo mismark	60%	3%	2%	75%	3%
Gross/net bias	70%	71%	80%	86%	88%
Mean of the six Totem anomalies	57%	64%	69%	73%	73%

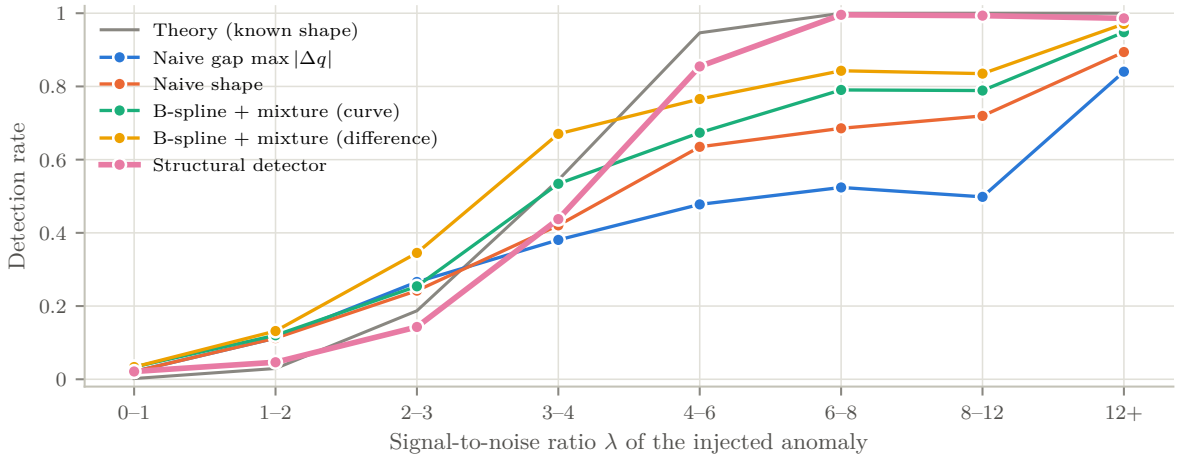


Figure 7: Detection rate as a function of the signal-to-noise ratio λ of the injected anomaly, over the six Totem anomalies. The grey curve is the theoretical power of a known typical shape at the typical threshold (Corollary 5.5).

Fig. 7 shows that the misses have a low signal-to-noise ratio.

- **Ex-date disagreements** are detected in 71% of cases; the naive shape check (74%) is competitive, because these spikes are large in repo.
- **The biased long sub-panel** escapes every method (24% for the structural detector, 42% for the naive gap): the bias builds up smoothly and blends into a repo slope. This is Theorem 3.6, not a tuning problem.
- **Genuine FO mismarks** are not taken for bad Totem curves: 3% alarms (the clean rate), and 80% are flagged as possible FO mismarks, against 2% of clean names. The naive gap and the mixture on the difference raise a bad-curve alarm on 60% and 75% of them (Proposition 6.2). The mismark test finds 97% of the errors on European stocks and 100% on the index, but only 24% on hard-to-borrow names: contributors' repo views there differ by about 60 bp, so an FO difference of 15 to 50 bp stays within the uncertainty of the consensus itself.

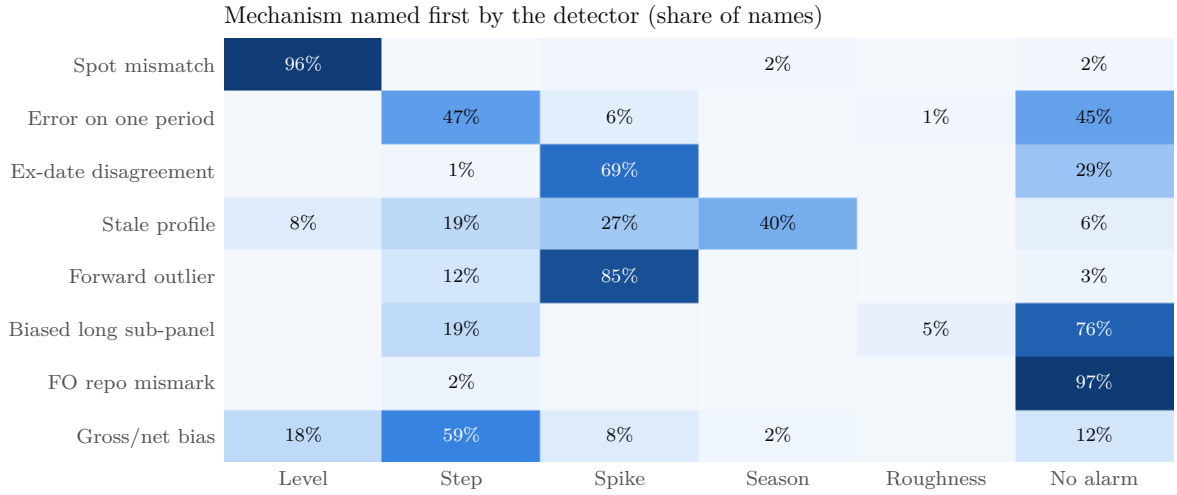


Figure 8: Mechanism named first by the structural detector, per scenario. Level, step, spike and season are the families of typical shapes; "roughness" means only the global test fired; "no alarm" that the name was not flagged.

- **The gross/net bias** is flagged in 88% of cases, mostly as steps at the ex-dates of discrete payers; for an index dividend stream it looks like a repo slope (Proposition 3.5, (vi)).
- **Power.** For a signal-to-noise ratio between 4 and 6, the structural detector finds 85% of the anomalies, and above 6 from 100% to 99%, in line with Corollary 5.5. Below 3, the naive gap and the mixture on the difference detect more (35% against 14% between 2 and 3): they also react to the part of an anomaly that looks like a smooth repo difference, which the structural detector deliberately leaves to the FO mismark test. The same property makes them flag genuine FO mismarks.
- **Explanation.** When the structural detector flags a Totem anomaly, its first explanation names the injected mechanism and its position in 88% of cases.

Results by category, areas under the ROC curve and realised false alarm rates are in Appendix C.2.

6.3 What becomes of the starting ideas

B-spline plus Gaussian mixture. The starting idea summarised each curve by its B-spline coefficients, by region and category, and clustered these coefficients with a Gaussian mixture. Two properties limit this approach.

Proposition 6.1 (What the coefficients do not see)**.** *Let $Z \in \mathcal{M}_{N,n_s}(\mathbb{R})$ be the matrix of the n_s B-spline functions at the pillars and $H = Z(Z^\top Z)^{-1}Z^\top$ the associated projection matrix. The least-squares fit of a unit spike e_k has energy $\|He_k\|^2 = h_{kk}$, with $\sum_k h_{kk} = n_s$. A fraction $1 - h_{kk}$ of the spike's energy stays in the residual and escapes every feature computed from the coefficients.*

Proof. H is symmetric and idempotent, so $\|He_k\|^2 = {e_k}^\top He_k = h_{kk}$, and $\sum_k h_{kk} = \operatorname{tr} H = n_s$. The coefficients $(Z^\top Z)^{-1}Z^\top e_k$ determine He_k and nothing else. □

For the benchmark's six cubic B-splines and eleven pillars:

Pillar	1M	2M	3M	6M	9M	1Y	18M	2Y	3Y	4Y	5Y
h_{kk}	1.00	0.80	0.48	0.58	0.32	0.38	0.37	0.34	0.51	0.37	0.86
$1 - h_{kk}$	0.00	0.20	0.52	0.42	0.68	0.62	0.63	0.66	0.49	0.63	0.14

At interior pillars, the coefficients capture only 0.32 to 0.80 of a spike's energy, and the spline spreads what it captures over several coefficients. A single-pillar anomaly, the most frequent pattern, is diluted before the clustering even starts.

Proposition 6.2 (Why a mixture on the difference flags FO errors)**.** *Take as features the coefficients $c = Z^+(q^{\text{tot}} - q^{\text{fo}})$, where Z^+ is the pseudo-inverse of Z. A parallel shift u of the FO repo moves them by $-u\,Z^+\mathbf{1}$, a fixed direction. If the mixture is fitted on clean history, where Totem and the FO agree up to noise, its density decreases along this direction: its alarm rate on FO mismarks tends to 1 as $|u|$ grows, whatever the quality of the Totem curve.*

Proof. c is linear in $q^{\text{tot}} - q^{\text{fo}}$, which moves by $-u\mathbf{1}$. Every Gaussian component has finite variance along $Z^+\mathbf{1}$: the mixture density, and with it the likelihood score, goes to zero along this direction. □

The structural detector does not have this defect, because $X\beta$ contains the level difference: it is removed from every test and published separately. The benchmark confirms both statements (75% against 3%).

Small neural networks per anomaly type.

Proposition 6.3 (A network can only learn the matched filter)**.** *After whitening and projection, $\hat{\varepsilon} \sim \mathcal{N}(0, \Pi)$ under H_0 and $\hat{\varepsilon} \sim \mathcal{N}(a\tilde{g}, \Pi)$ under H_1, for a known shape g and size a. Then*

$$\ln \frac{\mathbb{P}(H_1 \mid \hat{\varepsilon})}{\mathbb{P}(H_0 \mid \hat{\varepsilon})} = a\langle \tilde{g}, \hat{\varepsilon} \rangle - \frac{a^2}{2}\|\tilde{g}\|^2 + \ln \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)}, \tag{14}$$

an affine function of the matched filter $\langle \tilde{g}, \hat{\varepsilon} \rangle$. A classifier trained by cross-entropy on enough data converges to this posterior probability; with an unknown size and sign, one recovers the statistic $|z_g|$ of Theorem 5.4.

Proof. The two Gaussian distributions have the same covariance: the quadratic terms of their ratio cancel. Cross-entropy is minimised by the true posterior probability; maximising (14) over a gives $\langle \tilde{g}, \hat{\varepsilon} \rangle^2 / (2\|\tilde{g}\|^2)$. □

For the mechanisms one can name, a network can therefore at best learn the matched filter, which the structural detector writes down directly, with an exact false alarm rate and a known power.

Where these tools remain useful.

- **Penalised splines in the null model**, if $\beta_0 + \beta_1 \ln T$ turns out too rigid on real data.
- **Gaussian mixtures on the detector's outputs** ($\hat{\beta}_0$, $\hat{\beta}_1$, the z_g, $\|\hat{\varepsilon}\|^2$, the dispersion ratios), to learn the usual regimes by region and category in a low-dimensional space where every coordinate has a meaning.
- **Region × category** to pool the calibration of κ and of the level scale.
- **Neural networks**, once there is a history of labelled incidents and patterns the typical shapes do not cover; they should then be trained on the residuals $\hat{\varepsilon}$ and on synthetic injections, not on raw curves.

6.4 Limits

Limits

The synthetic market reproduces the mechanisms by construction; it does not show that real Totem data contain only these. Real data will also include stale contributors, corporate actions and special dividends, American exercise in single-stock quotes, and input errors. The scale κ absorbs a misspecified noise level, not a misspecified structure, and the injected sizes, though plausible, are not calibrated to any history. The benchmark shows that the method works when the mechanisms are present and is calibrated when they are absent; validation on twelve to twenty-four months of real history, with labelled incidents, is the next step.

Key points

- At equal false alarm rates, the structural detector detects as much as the best alternative on average, and clearly more on forward outliers.
- It does not take an FO mismark for a bad curve, unlike a Gaussian mixture on the Totem–FO difference, and it explains its alarms.
- B-splines, mixtures and neural networks remain useful, but downstream of the structural model, not on raw curves.

7 Recommendations and implementation

Goal of the section. Turn the results into control rules, and set out the data needed and the monthly procedure.

7.1 What we recommend

1. **Control the forward, not the split.** For each book, compute the forward line (Totem forwards, FO dividends) and the non-forward line $\langle \eta, \Delta\ell \rangle$. For forward-linear books, the forward line *is* the control, and it does not depend on the dividend contributors (Theorems 4.4 and 4.5). The Totem dividends are needed only where $\eta \neq 0$: dividend swaps and futures, stock borrow and lending, total return swap funding, American exercise, exotics.
2. **Do not add up without offsetting** large dividend and repo lines of opposite signs: this is a split artefact. In prudent valuation, take the forward as the parameter where $\eta \approx 0$.
3. **Judge the curve in carry, against a noise band.** Replace visual checks and thresholds in basis points of repo by the carry difference δ and the covariance Γ computed from Totem's dispersions and contributor counts (Proposition 5.2).
4. **Deploy the structural detector** with $\alpha = 2\%$, and publish for each alarm the explanation in natural units, the repaired curve and, separately, the FO mismark test.
5. **Keep B-splines, mixtures and neural networks downstream**: on the detector's outputs and residuals, and to pool the calibration by region and category.
6. **Calibrate and validate on real history**: κ and the level scale by region and category, every quarter, then a validation on twelve to twenty-four months of snapshots with

labelled incidents.

7.2 Data needed

Data	Fields	Size
Totem options	maturities; strikes; consensus call and put prices; contributor count; dispersion of the contributors' forwards (or an estimate); reference spot	N maturities
Totem dividends	period bounds; consensus level; dispersion; contributor count	m periods
FO market	spot at the Totem snapshot time; rate curve; repo at the Totem maturities; dividend forecast (dates and amounts)	N; list
FO risk	repo sensitivities per pillar, mapped onto the Totem maturities; dividend sensitivities per Totem period	$N + m$ per book
History	twelve to twenty-four monthly snapshots and known incidents	calibration, validation

If Totem does not publish the dispersion of the contributors' forwards, it can be estimated from the dispersion of deep in-the-money calls and puts, or calibrated on history: the detector needs the right shape across maturities, and κ corrects a common bias.

7.3 Monthly procedure

1. Load the Totem and FO snapshots at the same valuation time; check the reference spot.
2. Compute the forwards by parity; review the implied discount factor, the parity residuals and the dispersion ratios.
3. Allocate the Totem dividend levels on the FO ex-date profile; compute the implied repo with the Totem dividends (q^{tot}) and with the FO dividends ($\hat{q}$).
4. Build Γ and run the detector with the κ and level scale of the region and category.
5. Compute the control lines per book (forward, dividend, repo, non-forward) and the materiality of the split.
6. Review the alarms with their explanations, and the possible FO mismarks; use the repaired curve when a Totem anomaly is confirmed.
7. Record the decision taken on each alarm: these labels feed the next calibration and the validation.

7.4 Governance and next steps

κ and the level scale are recalibrated every quarter by region and category on clean history, with an alarm budget of 2% of names per month. The realised alarm rate and the share of explanations accepted by the controllers are tracked. Where they exist, market anchors complement the detector: listed index and dividend futures, total return futures, stock borrow fees for the short end. The next step is to run the pipeline on twelve to twenty-four months of real Totem and FO snapshots, calibrate the settings by region and category, and measure the quality of the explanations on labelled incidents.

Conclusion

Implied repo curves are irregular because they are the residual of two consensus averages computed on different panels, and part of that residual cannot be removed by any method

that only sees the consensus (Theorem 3.6). Two practical consequences follow. For control, a forward-linear book does not need the split between repo and dividends: the repo control is the forward control minus the dividend control, exactly under full revaluation, and the non-forward exposure η tells, book by book, when the split matters. For detection, the mechanisms that cause real anomalies have known shapes in the carry difference; matched filters, with a noise computed from Totem's own statistics, detect them at a fixed false alarm rate, explain them in natural units and tell them apart from FO mismarks.

A Proofs and complements

A.1 The bootstrap

Fix $q_1, \dots, q_{k-1}$ and set $x = q_k T_k$. For a dividend going ex in $(T_{k-1}, T_k]$, the interpolated cumulative repo is $q(t_i)\, t_i = q_{k-1} T_{k-1} + w_i\,(x - q_{k-1} T_{k-1})$, with $w_i = (t_i - T_{k-1})/(T_k - T_{k-1}) \in (0, 1]$. The model forward at T_k is therefore

$$\mathcal{F}_k(x) = S_0\, e^{r_k T_k - x} - \sum_{t_i \le T_{k-1}} d_i\, e^{(r_k T_k - r(t_i) t_i) - (x - q(t_i) t_i)} - \sum_{T_{k-1} < t_i \le T_k} d_i\, e^{(r_k T_k - r(t_i) t_i) - (1 - w_i)(x - q_{k-1} T_{k-1})}. \quad (15)$$

Lemma A.1 (Monotonicity)**.** *For every x, $\mathcal{F}_k'(x) \le -\mathcal{F}_k(x)$. In particular, the function $x \mapsto \ln \mathcal{F}_k(x) - \ln F_k$ has a derivative at most -1 where $\mathcal{F}_k > 0$.*

Proof. Differentiating (15) term by term,

$$\mathcal{F}_k'(x) = -S_0 e^{(\cdot)} + \sum_{t_i \le T_{k-1}} d_i e^{(\cdot)} + \sum_{T_{k-1} < t_i \le T_k} (1 - w_i)\, d_i e^{(\cdot)},$$

where each exponential is that of the matching term of (15). Every dividend term is positive and $1 - w_i \le 1$; replacing $1 - w_i$ by 1 can only increase the right-hand side, which then equals $-\mathcal{F}_k(x)$. □

Proof of Proposition 2.10. As $x \to -\infty$,

$$e^x\, \mathcal{F}_k(x) \longrightarrow e^{r_k T_k} \Big(S_0 - \sum_{t_i \le T_{k-1}} d_i\, e^{-(r(t_i) - q(t_i))\, t_i} \Big) = e^{r_k T_k} (S_0 - D_{k-1}) > 0,$$

because the terms of the current period only grow like $e^{-(1-w_i)x}$; so $\mathcal{F}_k(x) \to +\infty$. As x grows, $\mathcal{F}_k$ decreases strictly while it is positive (Lemma A.1), then tends to 0 or changes sign; by continuity, it takes every value in $(0, +\infty)$ exactly once on the branch where it is positive. On that branch, $x \mapsto \ln \mathcal{F}_k(x) - \ln F_k$ is C^2 with a derivative of absolute value at least 1: Newton's method converges quadratically near the root, and the bisection safeguard (keep a bracket of the root and fall back to its mid-point when the Newton step leaves it) makes the convergence global. □

Example A.2 (A bootstrap by hand)**.** Spot 100, flat rate 3%, dividends of 1.5 at $t = 0.30$ and 2.0 at $t = 0.80$, pillars 0.25, 0.5 and 1, true repo of 80, 70 and 60 bp. The forwards and the implied repo are:

Pillar T_k	Forward F_k	True q_k (bp)	Implied q_k (bp)
0.25	100.5515	80.00	80.00
0.50	99.6494	70.00	70.00
1.00	98.8928	60.00	60.00

At the third pillar, the Newton residual $|\ln \mathcal{F}_3(x) - \ln F_3|$ goes from 3.1×10^{-5} to 3.5×10^{-12} in one iteration, then to zero at machine precision at the next. With no dividend in the period, the function is affine and one iteration is enough.

A.2 Put–call parity

Lemma A.3 (Heterogeneous discount factors)**.** *If contributor j discounts with factor B_j, the zero of $\bar{C}-\bar{P}$ is $K^*=\sum_j B_jF_j/\sum_j B_j$, an average of the forwards weighted by the discount factors. Its distance to the simple average is $\mathrm{Cov}(B_j,F_j)/\bar{B}$, of second order.*

Proof. $\bar{C}-\bar{P}=\frac{1}{n}\sum_j B_j(F_j-K)$ vanishes at K^*, and $K^*-\bar{F}=(\overline{BF}-\bar{B}\,\bar{F})/\bar{B}$. □

Example A.4 (Parity)**.** Strikes 90, 100 and 110, puts at 2.1, 5.4 and 10.9, calls built from $F=102.3$ and $B(0,T)=0.97$. The weighted regression of $\bar{C}-\bar{P}$ on K gives $F=102.30$ and $B(0,T)=0.9700$. Multiplying every price by 1.02 (a wrong discount factor) gives $F=102.30$ and $B(0,T)=0.9894$: the forward does not change.

A.3 The noise covariance

Lemma A.5 (Cumulative covariance)**.** *Let $s\in\mathbb{R}^N$ have non-negative, non-decreasing components, and $w\in[0,1]$. The matrix $C_{kl}=w\,s_ks_l+(1-w)\min(s_k,s_l)^2$ is the covariance matrix of $Z_k=\sqrt{w}\,s_k\,\xi+\sqrt{1-w}\,B_{s_k^2}$, where $\xi\sim\mathcal{N}(0,1)$ is independent of a standard Brownian motion $(B_t)_{t\geq0}$; it is therefore symmetric positive semi-definite. For $w=0$, the correlation between Z_k and Z_l, $k<l$, is s_k/s_l.*

Proof. $\mathrm{Cov}(Z_k,Z_l)=w\,s_ks_l+(1-w)\,\mathrm{Cov}(B_{s_k^2},B_{s_l^2})=w\,s_ks_l+(1-w)\min(s_k^2,s_l^2)$. For $w=0$ and $k<l$, $\mathrm{Cov}(Z_k,Z_l)/\sqrt{\mathrm{Var}\,Z_k\,\mathrm{Var}\,Z_l}=s_k^2/(s_ks_l)=s_k/s_l$. □

Proof of Proposition 5.2. By Lemma 2.2, the error of contributor j on $\ln F_j(T_k)$ is $-T_ku_j-v_j(T_k)/(S_0-D_k)$. The repo part is a persistent level: its covariance between T_k and T_l is $\sigma_q^2T_kT_l$. The dividend part is cumulative, made of a persistent factor and increments at each ex-date: its covariance is that of Lemma A.5, applied to the dispersion s' (made non-decreasing). The two parts being independent, they add up; Proposition 3.4 gives the division by $\max(n_k^{\mathrm{opt}},n_l^{\mathrm{opt}})$, hence (11). For the dividend panel, Lemma 2.9 gives $e_k^{\mathrm{div}}=\sum_p a_{kp}\,\delta\ell_p/(S_0-D_k)=(A_S\,\delta\ell)_k$, where $\delta\ell$ is the error of the consensus levels, with covariance Γ^ℓ by Proposition 3.4; hence (12). The two panels being independent, $\Gamma=\Gamma^{\mathrm{opt}}+\Gamma^{\mathrm{div}}$. □

A.4 The Student distribution of the FO mismark test

Lemma A.6. *Let $\hat{\beta}_0\sim\mathcal{N}(\beta_0,c\,\sigma^2)$ and let $\hat{\sigma}^2$ be an independent estimate with $\nu\,\hat{\sigma}^2/\sigma^2\sim\chi^2(\nu)$. Then $(\hat{\beta}_0-\beta_0)/\sqrt{c\,\hat{\sigma}^2}$ follows a Student distribution with ν degrees of freedom.*

Proof. This is the definition of the Student distribution: a standard normal variable divided by the square root of an independent $\chi^2(\nu)/\nu$ variable. □

With dispersions estimated from n^{opt} contributors, $\nu\approx n^{\mathrm{opt}}-1$ at the short end, where the uncertainty on the repo level is read; the level scale s corrects what this approximation misses.

A.5 The second-order term of the control identity

Proposition A.7. *For a single pillar, with $F(q,D)=e^{(r-q)T}(S_0-D)$ and $V=\varphi\circ F$, the first-order total (dividend and repo lines) is*

$$\varphi'(F)\,\Delta F-\varphi'(F)\,F\Big[\tfrac{1}{2}T^2\,\Delta q^2+\frac{T\,\Delta q\,\Delta D}{S_0-D}\Big]+O(\|\Delta\|^3),$$

while the exact control is $\varphi'(F)\,\Delta F+\frac{1}{2}\varphi''(F)\,\Delta F^2+O(\|\Delta\|^3)$. All the dependence of the first-order total on the dividend panel is therefore in the bracket.

Proof. Expand F to second order, with $\partial_q F = -TF$, $\partial_D F = -F/(S_0 - D)$, $\partial^2_{qq} F = T^2 F$, $\partial^2_{qD} F = TF/(S_0 - D)$ and $\partial^2_{DD} F = 0$: $\Delta F = \partial_q F\,\Delta q + \partial_D F\,\Delta D + \frac{1}{2} T^2 F\,\Delta q^2 + TF\,\Delta q\,\Delta D/(S_0 - D) + O(3)$. The first-order total is $\varphi'(F)(\partial_q F\,\Delta q + \partial_D F\,\Delta D)$; substitute. □

For several pillars, the same expression adds up pillar by pillar, plus second-order cross terms created by the interpolation of the carry between pillars.

A.6 The signal-to-noise ratio in a paired design

Lemma A.8. *If a scenario adds an effect $e \in \mathbb{R}^N$ to a name's carry difference without changing its noise (same draw of contributors), then $\delta^{scenario} - \delta^{clean} = e$, and the signal-to-noise ratio $\lambda = \|\Pi W e\|$ of Corollary 5.5 is known exactly. For a single typical shape, $e = a\,g$ and $\lambda = |a|\,\|\tilde{g}\|$.*

Proof. The noise cancels in the difference; for $e = a\,g$, $\Pi W e = a\,\tilde{g}$. □

B Algorithms

Algorithm 1 Implied repo bootstrap

Require: spot S_0, rates $r(\cdot)$, dividends (t_i, d_i), pillars $T_1 < \cdots < T_N$, forwards $F_1, \ldots, F_N$

for $k = 1, \ldots, N$ **do**
  build $\mathcal{F}_k$ from (15) with $q_1, \ldots, q_{k-1}$
  $x \leftarrow$ starting point: dividends of the period discounted at the previous repo
  repeat
    $h \leftarrow \ln \mathcal{F}_k(x) - \ln F_k$; update the bracket $[x^-, x^+]$ from the sign of h
    $x \leftarrow x - h\,\mathcal{F}_k(x)/\mathcal{F}'_k(x)$; if $x \notin (x^-, x^+)$, $x \leftarrow (x^- + x^+)/2$
  until the step is below tolerance
  $q_k \leftarrow x/T_k$
return $(q_1, \ldots, q_N)$

Algorithm 2 Structural detection

Require: carry difference δ, covariance Γ, scale κ, matrix X, typical shapes $\mathcal{G}$, level α

$W \leftarrow L^{-1}$ with $\kappa\Gamma = L\,L^\top$; $\Pi \leftarrow$ projection onto the orthogonal complement of WX; $\hat{\varepsilon} \leftarrow \Pi W \delta$
for each $g \in \mathcal{G}$: $\tilde{g} \leftarrow \Pi W g$, $z_g \leftarrow \langle \tilde{g}, \hat{\varepsilon} \rangle / \|\tilde{g}\|$
$c_\alpha \leftarrow 1 - \alpha/2$ quantile of $\max_g |Z_g|$, $Z \sim \mathcal{N}(0, \Omega)$ ▷ Proposition 5.6
alarm $\leftarrow \max_g |z_g| > c_\alpha$ **or** $\|\hat{\varepsilon}\|^2 > \chi^2_{1-\alpha/2}(N-2)$
if alarm **then**
  $\mathcal{S} \leftarrow \emptyset$; **repeat**: add the shape with the largest $|z|$ after projecting out $[X, \mathcal{S}]$, **while** that $|z|$ exceeds c_α
joint generalised least-squares fit of δ on $[X, \mathcal{S}]$; sizes in natural units; repaired curve
possible FO mismark if $|\hat{\beta}_0| > \max(10\,\text{bp}, t_{\nu,1-\alpha/2}\,s\,\sigma(\hat{\beta}_0))$

Algorithm 3 Calibration of the scale κ

Require: clean names $i \in \{1, \dots, n\}$ with $(\max_g |z_{g,i}|, c_{\alpha,i}, \|\hat{\varepsilon}_i\|^2)$ computed for $\kappa = 1$; target α

$u^- \leftarrow \ln 10^{-3}$, $u^+ \leftarrow \ln 10^{3}$

for 60 iterations **do**

  $u \leftarrow (u^- + u^+)/2$; $\hat{\alpha} \leftarrow \frac{1}{n} \sum_i \mathbf{1}\{\max_g |z_{g,i}|\, e^{-u/2} > c_{\alpha,i} \text{ or } \|\hat{\varepsilon}_i\|^2 e^{-u} > \chi^2_{1-\alpha/2}(N-2)\}$

  if $\hat{\alpha} \le \alpha$ **then** $u^+ \leftarrow u$ **else** $u^- \leftarrow u$

return $\kappa = e^{u^+}$

C Benchmark: specification and further results

C.1 Specification of the synthetic market

- **Dividends.** European stocks: annual dividend going ex between 20 April and 30 May (35% of names also pay a 40% interim in November); US stocks: quarterly; hard-to-borrow names: annual or quarterly; European index: monthly stream with weights $(1, 2, 4, 14, 24, 18, 6, 4, 3, 4, 3, 2)$ per calendar month. Yield and growth drawn per name within the ranges of Table 4.
- **Repo.** $q(T) = q_\infty + (q_0 - q_\infty)\, e^{-T/\tau}$, with q_0, q_∞ and τ drawn per name.
- **Contributors' dividend views.** $d_{ij} = d_i(1 + \sigma(t_i)\,[0.7\, z_j + \sqrt{1 - 0.7^2}\, e_{ij}])$, with $\sigma(t)$ affine in t and specific to each category, z_j persistent and e_{ij} specific to each dividend; dividends already announced (before 0.15 years) have a dispersion ten times smaller.
- **Contributors' repo views.** $q_j(T) = q(T) + \sigma_q\big(u_j + 0.6\, v_j\, \varsigma(T)\big)$, where ς is a tilt linear in $\ln T$ from -0.5 to 0.5; σ_q is 15, 10, 60 and 8 bp for the four categories.
- **Options.** Strikes from 80% to 120% of spot; volatility with a term structure and a smile slope of -0.10 per unit of log-moneyness; volatility noise of one point per contributor, price noise of 2×10^{-5} of spot.
- **Consensus.** Panel mean after excluding quotes more than four normalised median absolute deviations away from the median.
- **FO.** Dividends with a 1% noise, repo with a 3 bp level error; spot equal to the Totem reference spot.

C.2 Further results

Table 7: Area under the ROC curve against clean names (ranking quality, independent of the threshold).

Scenario	Naive gap	Naive shape	Spline + mixture, curve	Spline + mixture, difference	Structural detector
Spot mismatch	0.98	0.85	1.00	1.00	1.00
Error on one period	0.70	0.72	0.81	0.86	0.84
Ex-date disagreement	0.77	0.87	0.84	0.87	0.89
Stale profile	0.96	0.96	0.97	0.97	0.97
Forward outlier	0.71	0.83	0.87	0.90	0.99
Biased long sub-panel	0.75	0.63	0.76	0.80	0.74
FO repo mismark	0.85	0.50	0.50	0.96	0.50
Gross/net bias	0.90	0.87	0.97	0.98	0.97

Table 8: False alarm rate realised on new clean names, by category (target 2%), and calibrated scale κ.

Category	Naive gap	Naive shape	Spline + mixture, curve	Spline + mixture, difference	Structural detector	κ
EU stock	4.0%	3.3%	1.3%	4.0%	3.3%	1.60
US stock	3.3%	4.0%	1.3%	1.3%	5.3%	1.88
Hard to borrow	1.3%	1.3%	1.3%	4.0%	3.3%	0.96
EU index	1.3%	2.0%	6.0%	2.0%	0.7%	3.58

Table 9: Structural detector: share of names flagged, by scenario and category.

Scenario	EU stock	US stock	Hard to borrow	EU index
Clean	3%	5%	3%	1%
Spot mismatch	100%	100%	94%	100%
Error on one period	51%	65%	27%	76%
Ex-date disagreement	81%	90%	57%	55%
Stale profile	100%	100%	95%	83%
Forward outlier	99%	99%	100%	91%
Biased long sub-panel	12%	17%	5%	63%
FO repo mismark	4%	5%	3%	1%
Gross/net bias	100%	100%	53%	100%

Table 10: Structural detector: detection, explanation (the first explanation names the injected mechanism and its position) and FO mismark flags.

Scenario	Flagged	Right mechanism (among flagged)	FO mismark flag
Spot mismatch	98%	97%	2%
Error on one period	55%	80%	7%
Ex-date disagreement	71%	97%	2%
Stale profile	94%	71%	23%
Forward outlier	97%	97%	2%
Biased long sub-panel	24%	66%	2%
FO repo mismark	3%	–	80%
Gross/net bias	88%	–	41%

D Notation

Symbol	Meaning
S_0, $r(T)$, $B(0,T)$	spot, zero rate, discount factor
$q(T)$, q_k	zero repo rate, its value at pillar T_k
t_i, d_i, $\tilde{d}_i$	ex-dates, dividends, dividends discounted at the carry rate
$D(T)$, $F(T)$	cumulative discounted dividends, forward (Definition 2.1)
$T_1 < \cdots < T_N$	Totem maturities (pillars), $N = 11$
I_p, ℓ_p, π_i	dividend periods, $p \in \{1, \ldots, m\}$, their levels, allocation profile weights
$A = (a_{kp})$, A_S	allocation matrix (Lemma 2.9), $(A_S)_{kp} = a_{kp}/(S_0 - D_k)$
$\mathcal{P}_k^{\text{opt}}$, $\mathcal{P}_p^{\text{div}}$	option and dividend panels, of sizes n_k^{opt} and n_p^{div}; n^{com} common contributors
$\bar{q}^{\,\text{opt}}$, $\bar{D}^{\text{opt}}$, $\bar{D}^{\text{div}}$	average repo and dividends of the option panel, dividends of the dividend panel
x^{tot}, x^{fo}	Totem quantity, FO quantity
δ_k	carry difference $(q_k^{\text{tot}} - q_k^{\text{fo}})\,T_k$ (Definition 2.3)
$\hat{q}$	repo implied by the Totem forwards with the FO dividends
ψ, J	implied repo map, its Jacobian $\partial_\ell \psi$ (Lemma 4.3)
s^q, s^ℓ, η	FO sensitivities to repo and dividends, non-forward exposure $s^\ell + J^\top s^q$
X, $\beta = (\beta_0, \beta_1)$	null model: smooth repo difference $\beta_0 + \beta_1 \ln T$
$\Gamma = \Gamma^{\text{opt}} + \Gamma^{\text{div}}$, Γ^ℓ, κ	covariance of the carry noise, of the dividend levels, scale
g, a_g, $\mathcal{G}$, $\mathcal{S}$	typical shape, its size, set of typical shapes, selected shapes
W, Π, $\hat{\varepsilon}$, $\tilde{g}$	whitening, projection, whitened residual, whitened typical shape
z_g, c_α, λ	matched-filter statistic, common threshold, signal-to-noise ratio

E Reproducibility

Every number, table and figure in this note is produced by the code in the repository:

```
pip install -e .[benchmark,report,dev]
pytest
python scripts/run_benchmark.py    (about 4 minutes)
PYTHONPATH=.  python scripts/build_latex.py
```

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